Monitoring and Predicting Floods Using Earth Observations for Planning and Preparedness

Part 1: Overview of Global Flood Product Derived from NASA Optical Observations

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Guest Speaker: Daniel Slayback (618 NASA-GSFC, SSAI)

June 18, 2026

About ARSET

About ARSET

- **ARSET provides accessible, relevant, and cost-free training on remote sensing satellites, sensors, methods, and tools.**
- Trainings include a variety of applications of satellite data and are tailored to audiences with a variety of experience levels.

AGRICULTURE

DISASTERS

ECOLOGICAL CONSERVATION

HEALTH & AIR QUALITY

WATER RESOURCES

WILDLAND FIRES

About ARSET Trainings

- Online or in-person
- Live and instructor-led or asynchronous and self-paced
- Cost-free
- Bilingual and multilingual options
- Only use open-source software and data
- Accommodate differing levels of expertise

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Monitoring and Predicting Floods Using Earth Observations for Planning and Preparedness
Overview

Importance of Flood Monitoring and Prediction

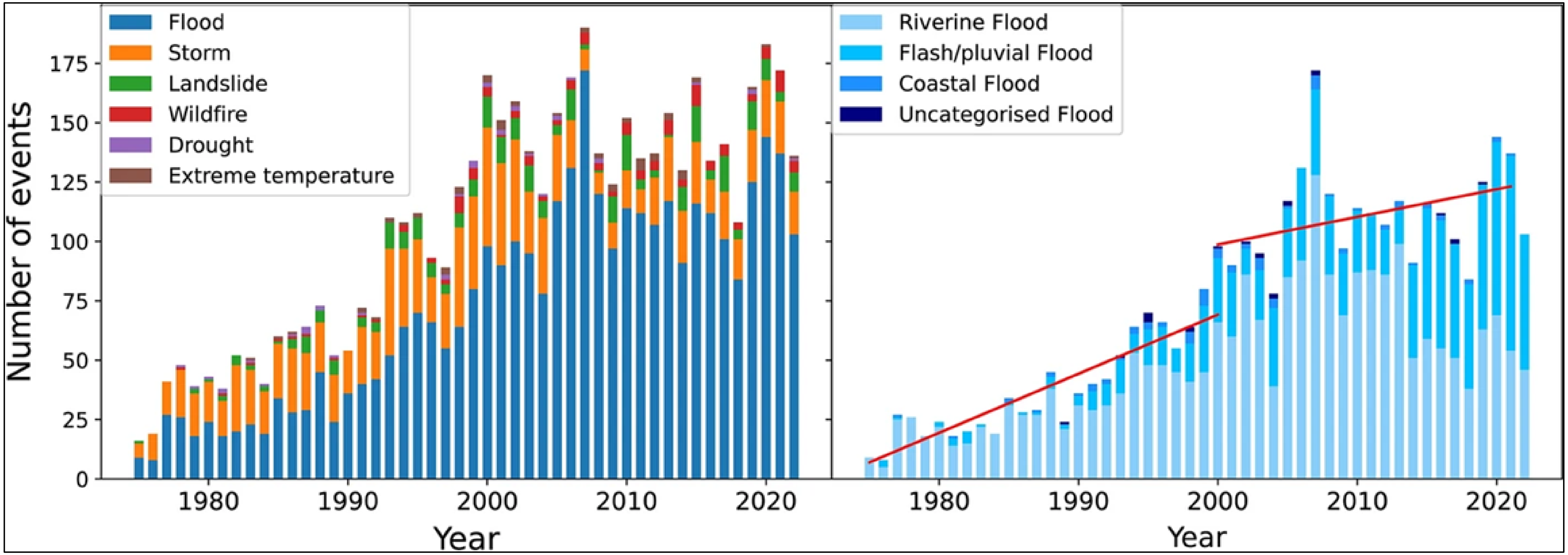
- Floods are the most common and widespread of all weather-related natural disasters.
- Floods last from a few minutes to weeks, placing a huge burden on communities:
  - Loss of lives
  - Displacement of communities
  - Long-term impacts on human health and well-being
  - Damage to infrastructure and economies
  - Destruction of ecosystems
- In 2025 China, India, Nepal, Pakistan, the Republic of Korea were among the countries affected in Asia, with hundreds of lives lost, whilst flash flooding in the US states of Texas and New Mexico also killed more than 100 people. (World Meteorological Organization – WMO)

Major flood disasters in Asia and the United States of America have caused heavy casualties and economic losses in July and the first days of August, highlighting the dangers and the challenges in ensuring that early warnings reach those who need them.

Image Source: WMO

Major floods occurred in Asia and United States in July & August 2025

Global Flood Events (1975-2022)



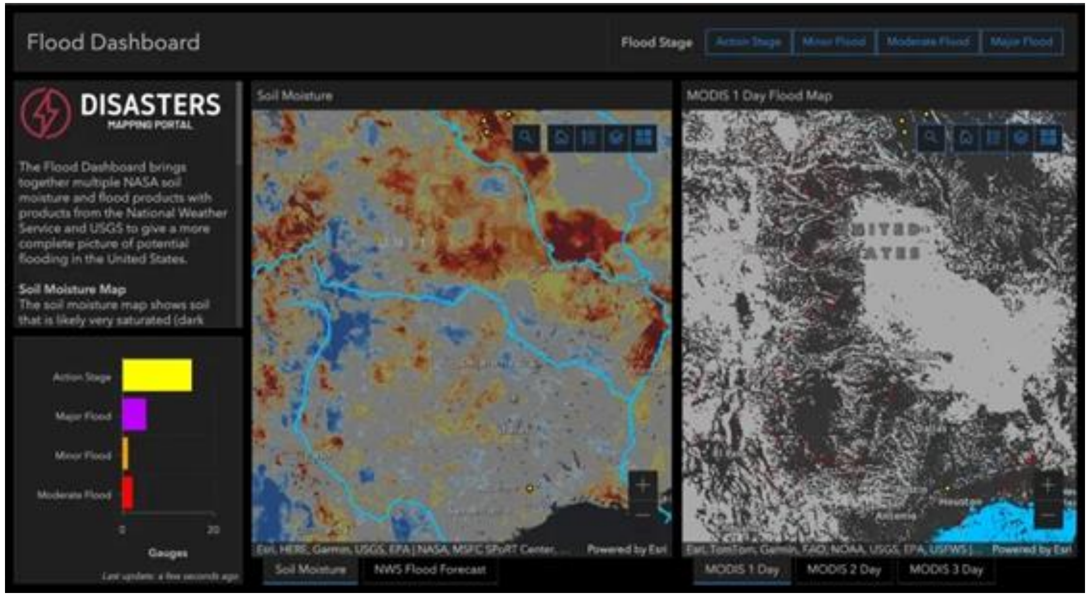
Jonkman et al. (2024)

Data Source:
EM-DAT: The International Disasters Database
Center for Research on the Epidemiology of Disasters

Remote Sensing Observations for Flood Applications and Impact

- Optical reflectances and Synthetic Aperture Radar (SAR) backscatter data are used for flood detection.
- When combined with hydrologic models, remote sensing observations, such as precipitation, soil moisture, digital elevation, and land cover can support both **flood monitoring and prediction efforts**.
  - early warning for enhanced safety
  - planning rescue and relief operations
  - planning Infrastructure and economic impact mitigation

NASA Flood Dashboard



Additional information can be found at NASA Earthdata/Floods and Mapping Flood Impacts.

Training Learning Objectives

By the end of this training attendees will be able to:

- Identify datasets in NASA Global Flood Product based on remotely sensed optical observations.
- Use the webtool, NASA Worldview, to access and visualize flooded regions from Global Flood Product, available from multiple satellites, from 2021 to near-real time.
- Identify OPERA dynamic surface water extent data for flood detection derived from optical and SAR observations.
- Access and visualize OPERA dynamic surface water extent data for flood events using NASA Worldview.
- Identify the capabilities of GEOGLOWS River Forecast System (RFS) for global streamflow prediction.
- Use GEOGLOWS Hydroviewer to access globally available retrospective and predicted streamflow for selected rivers.

OPERA: Observational Products for End-Users from Remote Sensing Analysis
GEOGLOWS: Group on Earth Observations (GEO) GLObal Water Sustainability

Prerequisites

- Fundamentals of Remote Sensing

Training Outline

Part 1	Part 2	Part 3
Overview of Global Flood Product Derived from NASA Optical Observations	Overview of Monitoring Floods using OPERA Surface Water Extent based on Optical and SAR Observations	Overview of GEOGLOWS for Monitoring and Predicting Flood Risk
June 18, 2026	June 23, 2026	June 25, 2026

Homework

Opens June 25 – Due July 09 – Posted on Training Webpage

A certificate of completion will be awarded to those who attend all live sessions and complete the homework assignment(s) before the given due date.

Part 1: Overview of Global Flood Product Derived from NASA Optical Observations

Part 1 Objectives

By the end of Part 1, participants will be able to:

- Identify NASA Global Flood Product derived from remotely sensed **optical observations**.
- Recognize the data sources, spatial/temporal resolution, and limitations of optical-based flood detection.
- Gain experience using NASA Worldview to access and visualize Global Flood Product in near real-time.
- Know where to access flood product files for download.

Part 1 Outline

- Examples of flood cases using Near-real Time (NRT) Global Flood Product
- History and background of Global Flood Product
- Overview of NRT Global Flood Product
  - Flood detection approach based on optical imagery from MODIS[1] on Terra and Aqua satellites and VIIRS[2] on NOAA-20 and -21 satellites
  - Flood product evaluation
  - Case studies of flood detection
- Demonstration: NRT Global Flood Products: access and visualization using NASA Worldview

[1]MODIS: MODerate-Resolution Imaging Spectroradiometer
[2]VIIRS: Visible and Infrared Imaging Radiometer Suite

How to Ask Questions

- Please put your questions in the Questions box and we will address them at the end of the webinar.
- Feel free to enter your questions as we go. We will try to get to all of the questions during the Q&A session after the webinar.
- The remainder of the questions will be answered in the Q&A document, which will be posted to the training website about a week after the training.

Part 1 – Trainer

Dr. Daniel Slayback

Research Scientist

618 NASA-GSFC

Overview of NASA's Global Flood Product Derived from Optical Observations

Dan Slayback[1,2]

Research Scientist

Flood Team: Diane Davies[1,2], Sadashiva Devadiga[1], Asen Radov[1,3], Ranjay Shrestha[1,2]

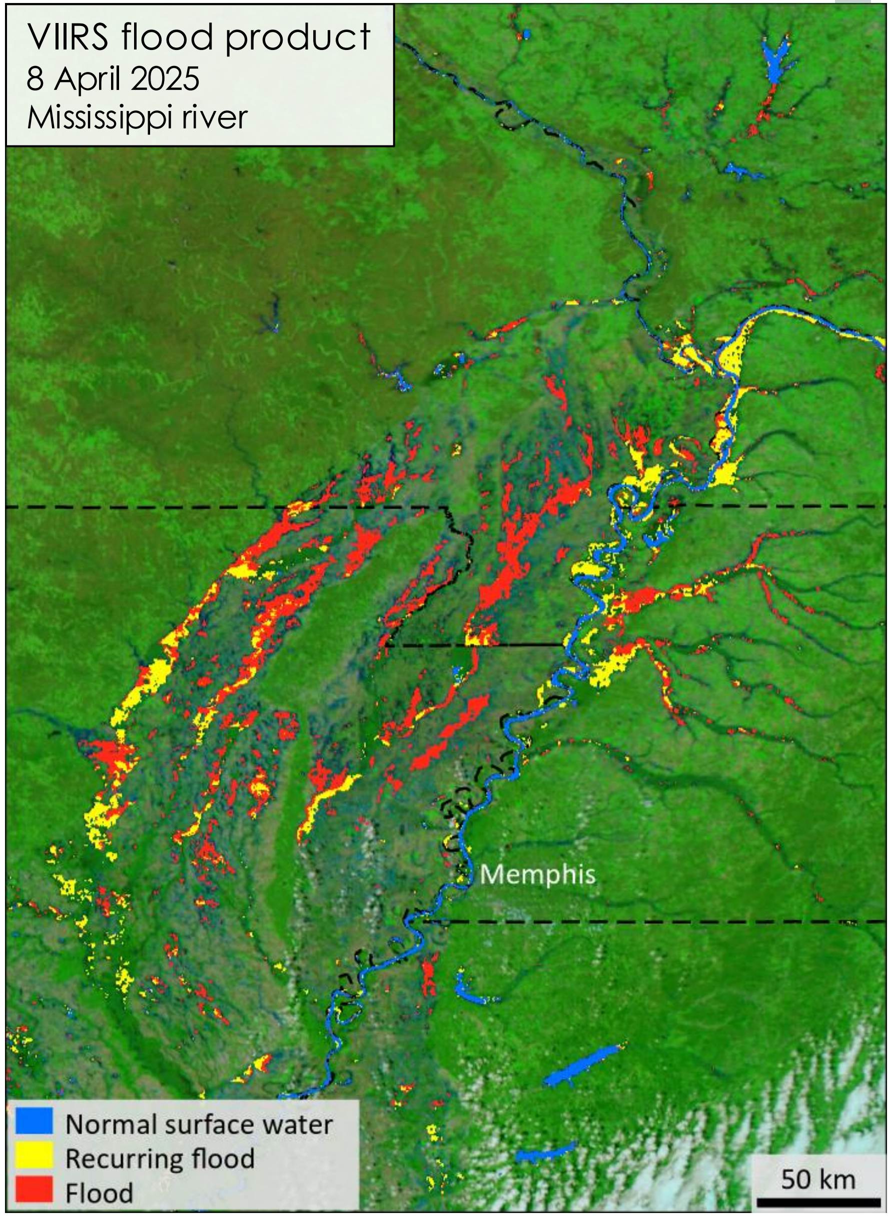
[1]NASA Goddard Space Flight Center

[2]Science Systems & Applications, Inc.

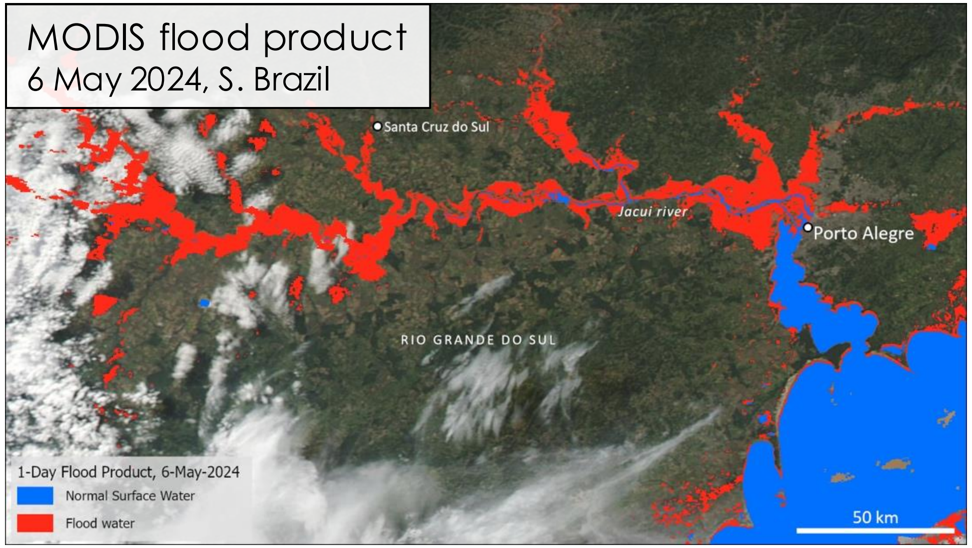
[3]University of Maryland, College Park

NASA's Global Flood Product: Talk Outline

- Product Overview & History
- Approach
- Recent Updates
- Evaluation
- Limitations
- Distribution
- Archive
- Future directions



Near Real-Time Global Flood Products: Product Overview & Brief History

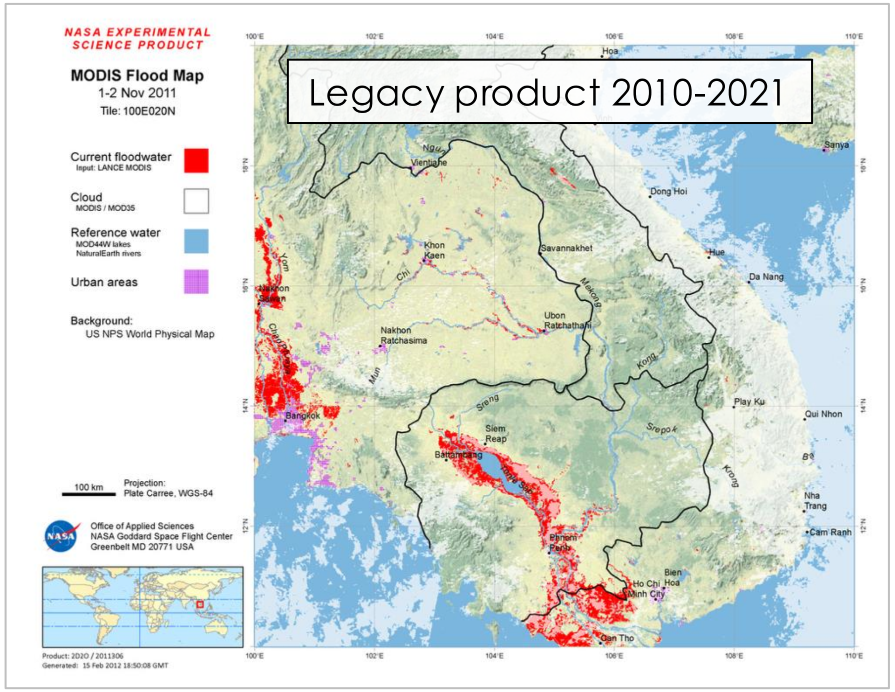


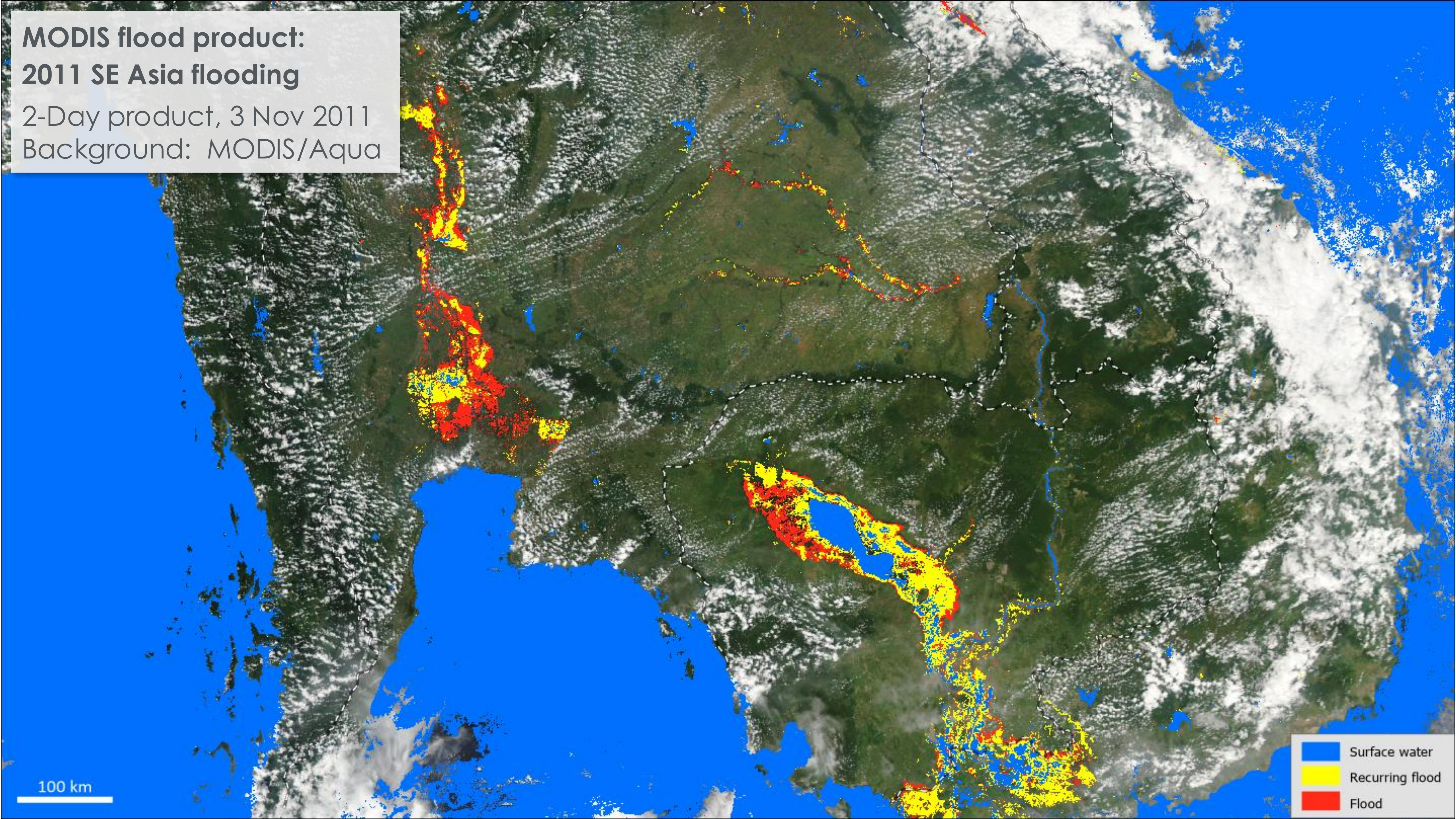
Key features

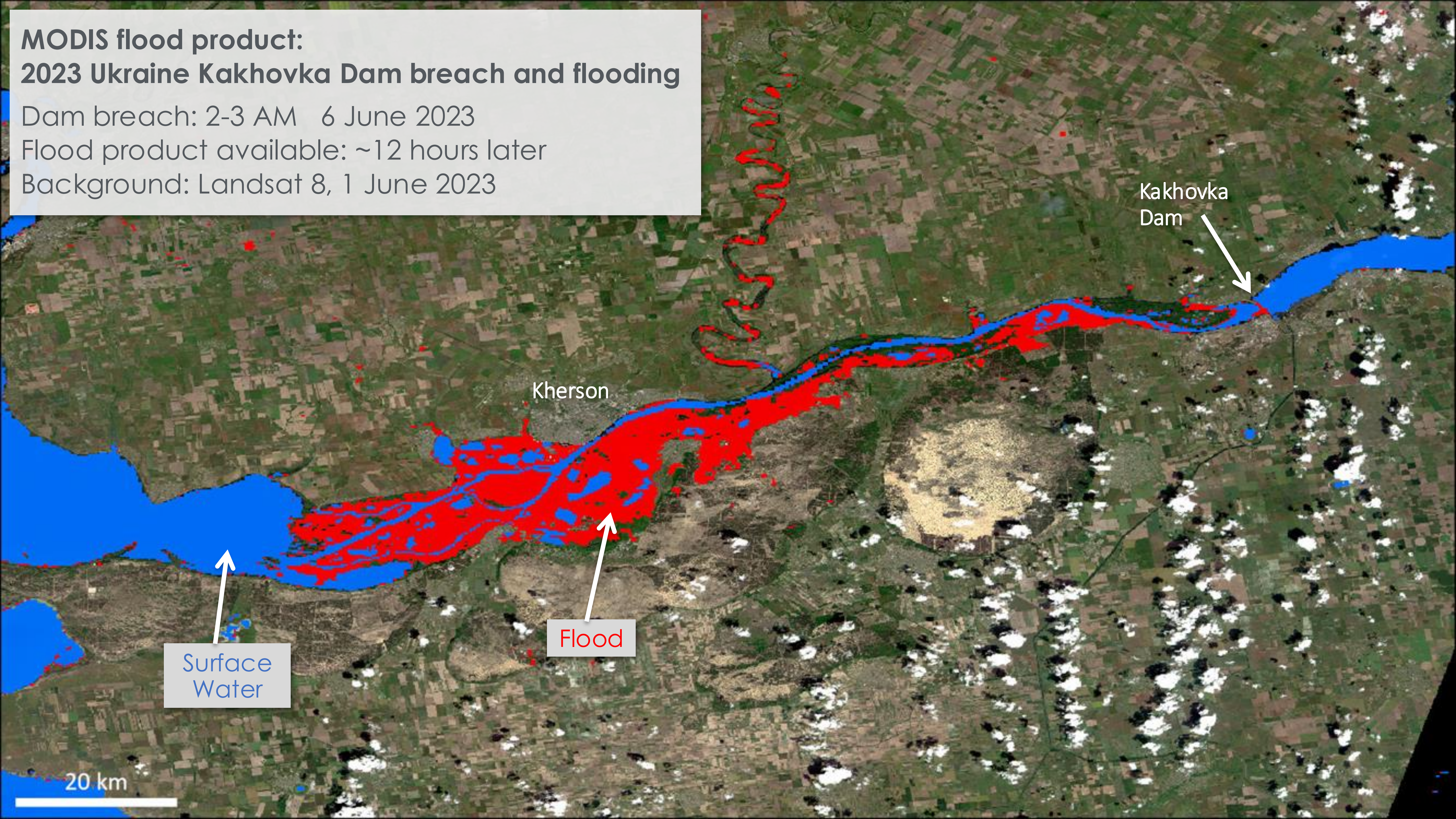
- Daily
- Global
- 250 m resolution
- Twice daily observations
- Near real-time (within 3 hours of satellite overpass)
- Detects non-flood and flood water; "flood" determined by comparing detected water to reference water layers
- Limitations: persistent cloud cover; flash flood/small floods

History

- 2005: Bob Brakenridge (Dartmouth Flood Observatory) manually generated flood maps from MODIS Rapid Response imagery
- 2010: NASA GSFC's Office of Applied Science initiated a project to automate production with LANCE inputs →
- 2021: Product fully transitioned to LANCE production systems
- 2025: VIIRS product launched (to eventually replace MODIS)



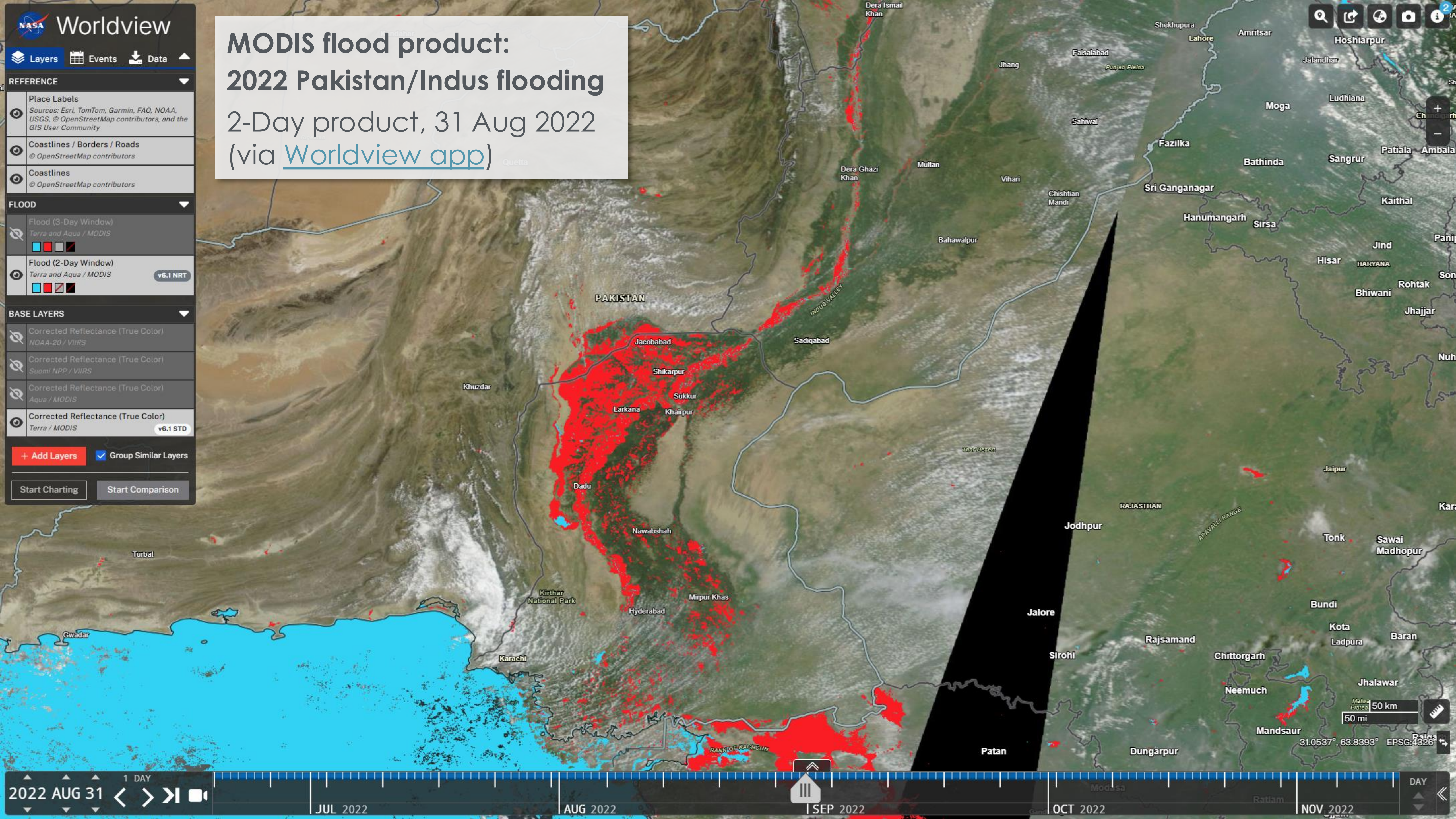

MODIS flood product:
2011 SE Asia flooding
2-Day product, 3 Nov 2011
Background: MODIS/Aqua
Surface water
Recurring flood
Flood
100 km



MODIS flood product:
2023 Ukraine Kakhovka Dam breach and flooding
Dam breach: 2-3 AM 6 June 2023
Flood product available: ~12 hours later
Background: Landsat 8, 1 June 2023
Kakhovka Dam
Kherson
Flood
Surface Water
20 km

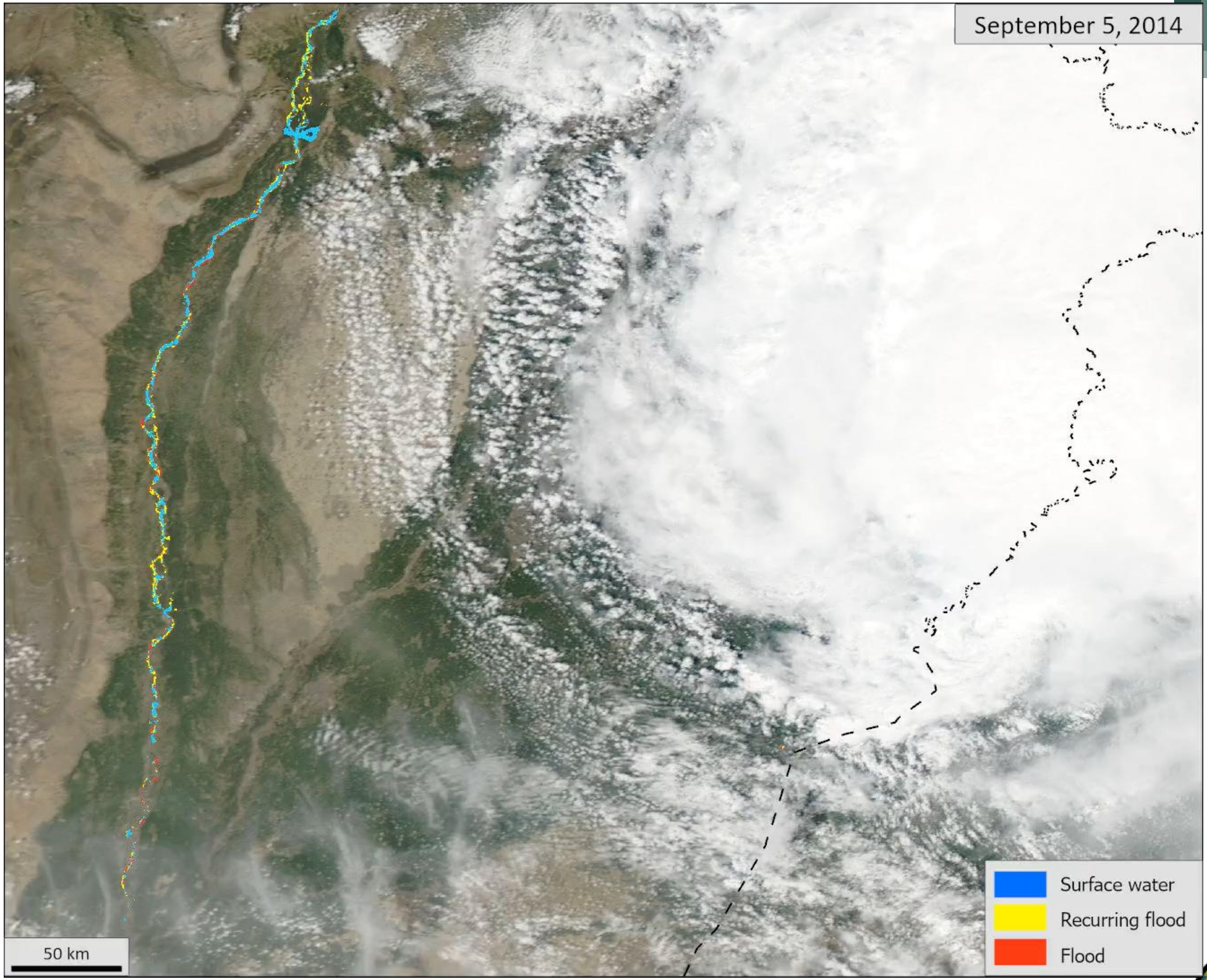
MODIS flood product: 2022 Pakistan/Indus flooding

2-Day product, 31 Aug 2022 (via Worldview app)



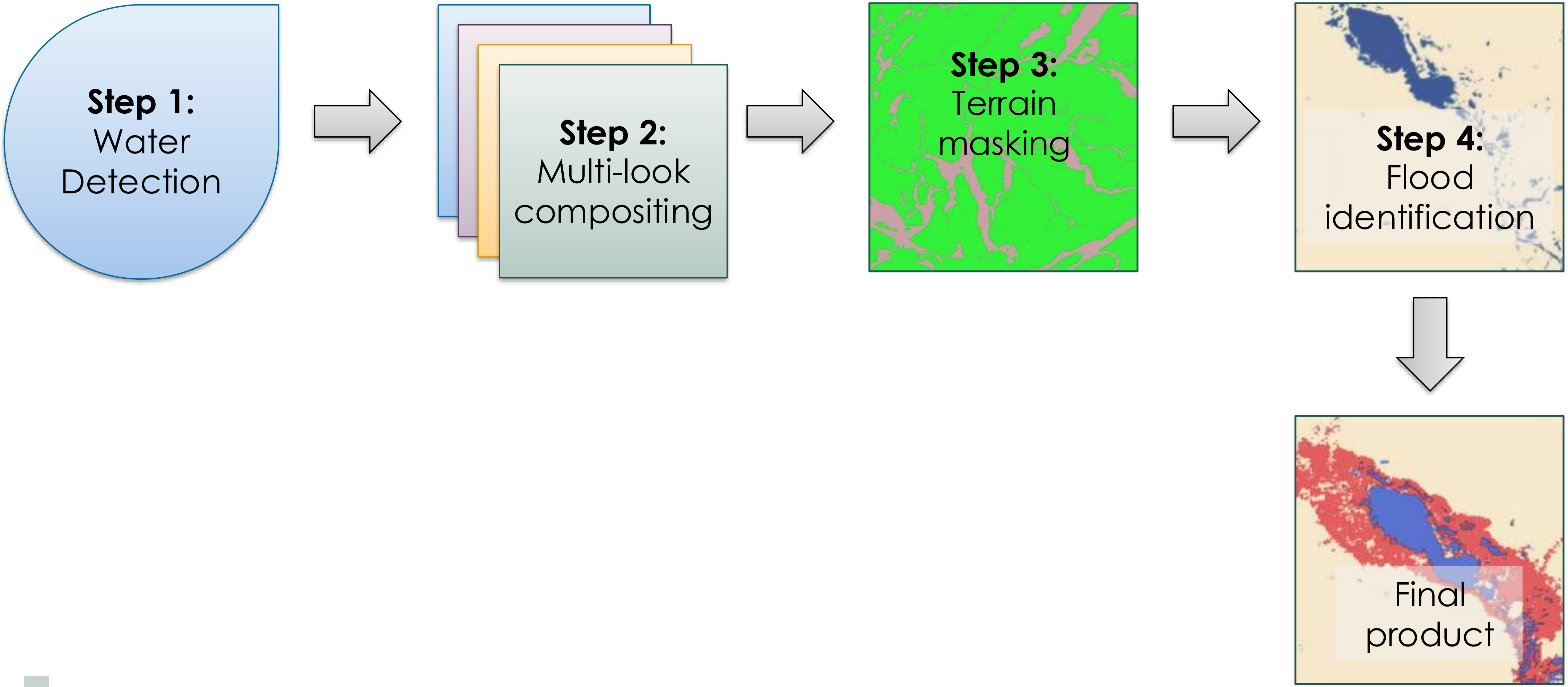
MODIS product time-series: Pakistan flooding 2014

2-day composite product
5 - 21 Sept 2014
Background: MODIS/Aqua



Flood Product Approach

Overview: Four Basic Steps



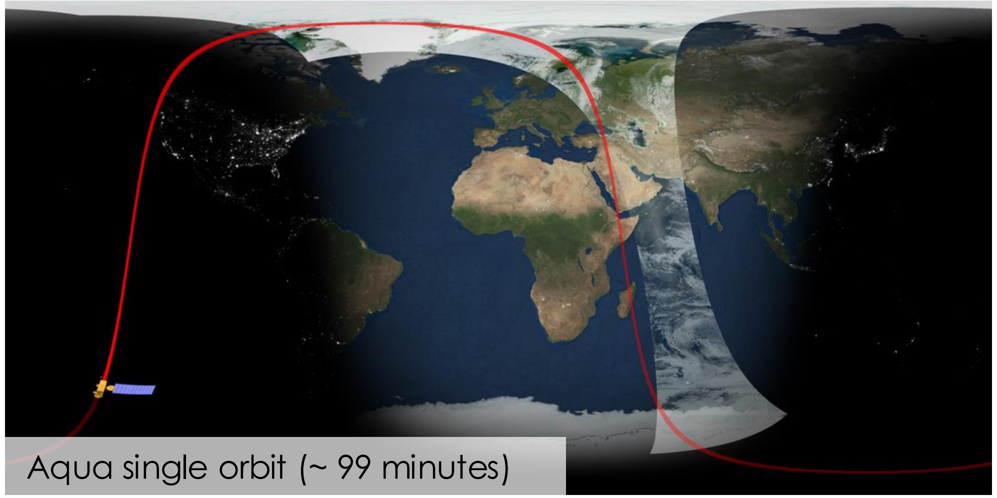
MODIS Product Source Imagery

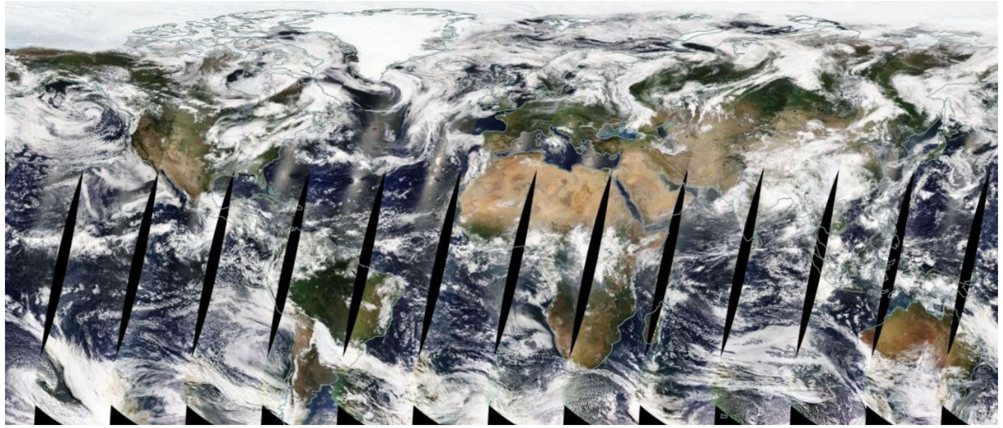
MODerate resolution Imaging Spectroradiometer

Step 1: Water Detection

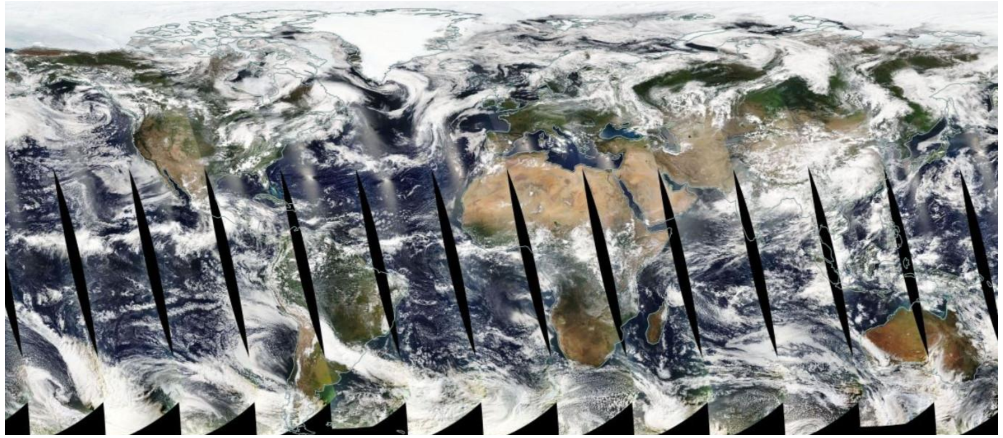
Key features:

- On board two NASA polar-orbiting satellites:
    - Terra: 1999 – present (ending ~2026/27)
    - Aqua: 2002 – present (ending ~2026/27)
- 250 m resolution
- Global
- 2 x Daily
    - Terra: ~10:30 AM local time
    - Aqua: ~1:30 PM local time
- Optical (cannot see through cloud)





MODIS / Terra (10:30 am) Daily Collect: 6/18/2025



MODIS / Aqua (1:30 pm) Daily Collect: 6/18/2025

VIIRS Product Source Imagery

Visible Infrared Imaging Radiometer Suite

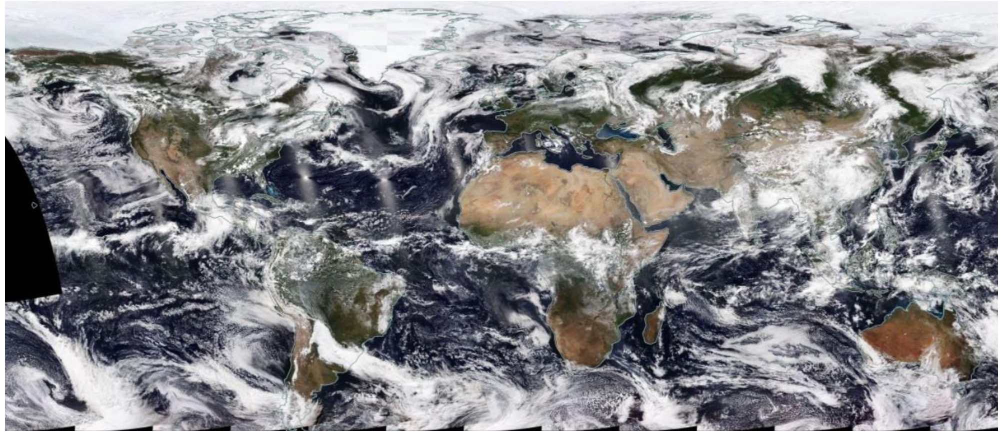
Step 1: Water Detection

Key features:

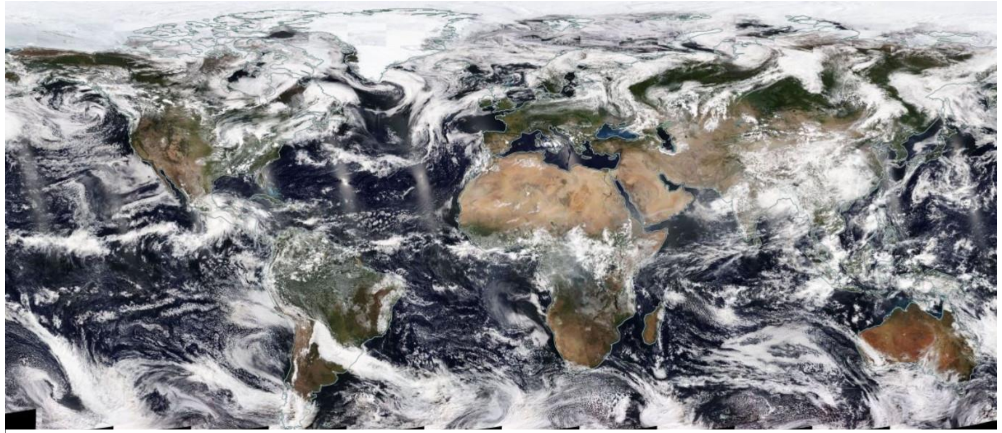
- On board two NOAA polar-orbiting satellites:
  - NOAA-20: 2017 – present
  - NOAA-21: 2022 – present
- 375 m resolution
- Global
- 2 x Daily
  - NOAA-20: ~1:30 PM local time
  - NOAA-21: ~2:20 PM local time
- Optical (cannot see through cloud)

Differences from MODIS:

- 375 m resolution, vs 250
- Only afternoon overpasses (within 50 minutes) vs 10:30 am / 1:30 pm (3-hour gap)
- No equatorial swath gaps!



VIIRS / NOAA-20 (1:30 pm) Daily Collect: 6/18/2025



VIIRS / NOAA-21 (2:20 pm) Daily Collect: 6/18/2025

Water Detection Algorithm

Step 1: Water Detection

Three conditions must be met:

$$\frac{(NearIR + A)}{(Red + B)} < C$$

AND $Red < D$

AND $SWIR < E$

A	13.5
B	1081
C	0.7
D	2027
E	676

Thresholds in units of scaled reflectance (0-10000)
Empirically developed by Bob Brakenridge, Dartmouth Flood Observatory

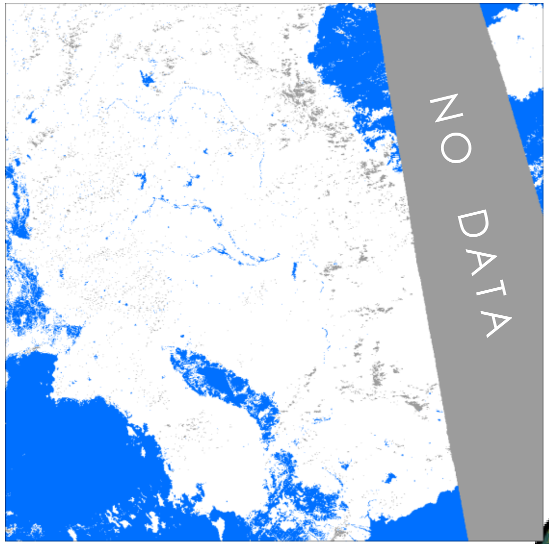
Band (Band #)	Wavelengths (nm)	
(MODIS, VIIRS)	MODIS	VIIRS
Red (1, i1)	620-670	600-680
Near-IR (2, i2)	841-876	850-880
SWIR (7, m11)	2105-2155	2230-2280

10° x 10° tile

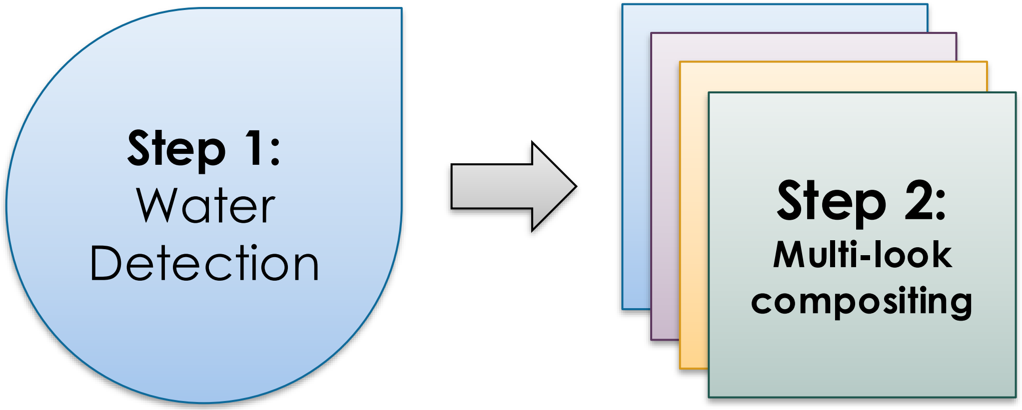
Input Image: MODIS/Aqua

→ Water detection algorithm →

Output: Detected water



Multi-look Compositing



Idea: extend input data over 1, 2, or 3 days to:

1. Help address false-negatives, with additional imagery where clouds may have moved exposing the surface
2. Help address cloud-shadow false-positives by requiring multiple water detections (in 2 and 3-day composites) to mark a pixel as water.

Approach:

1. Sum water detections from all available observations, over the composite period (1, 2, or 3 days).
2. If SUM >= THRESHOLD: mark pixel as water

The threshold is generally equal to half the number of observations:

- 1-day composite (2 obs): 1
- 2-day composite (4 obs): 2
- 3-day composite (6 obs): 3

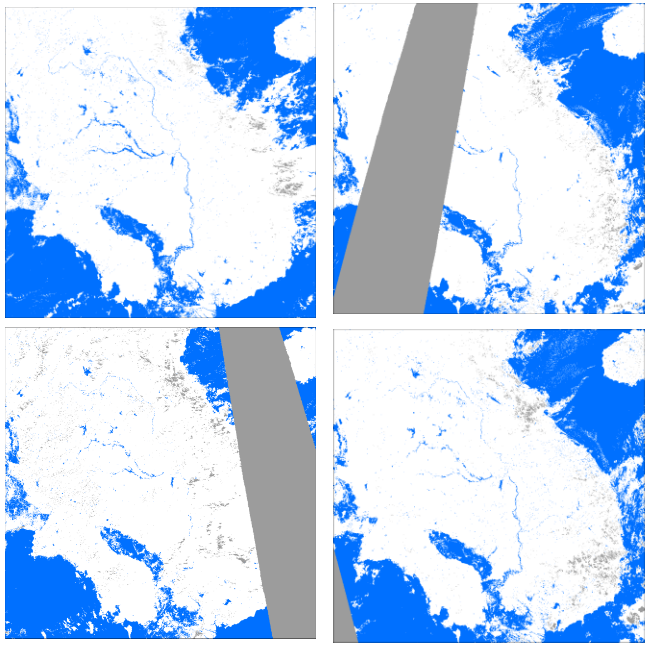
(modified by the actual number of observations)

2-day composite example

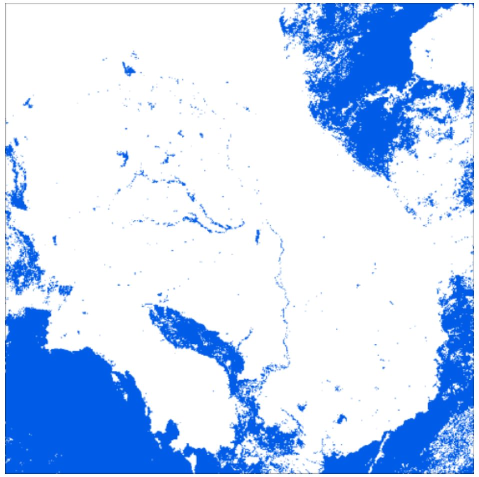
Day 1 | Day 2

MODIS/Terra 10:30 AM

MODIS/Aqua 1:30 PM



Composited water



Terrain Masking

Step 1: Water Detection

Step 2: Multi-look compositing

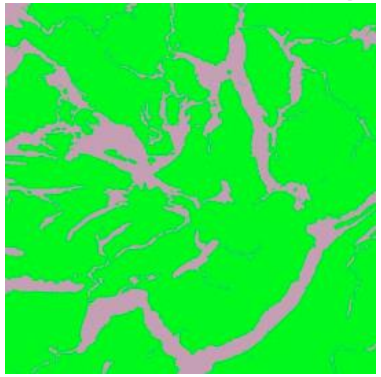
Apply two terrain masks to remove likely false-positive water detections

1. Monthly computed terrain shadow masks.

MODIS/Terra
February, 10:30 AM

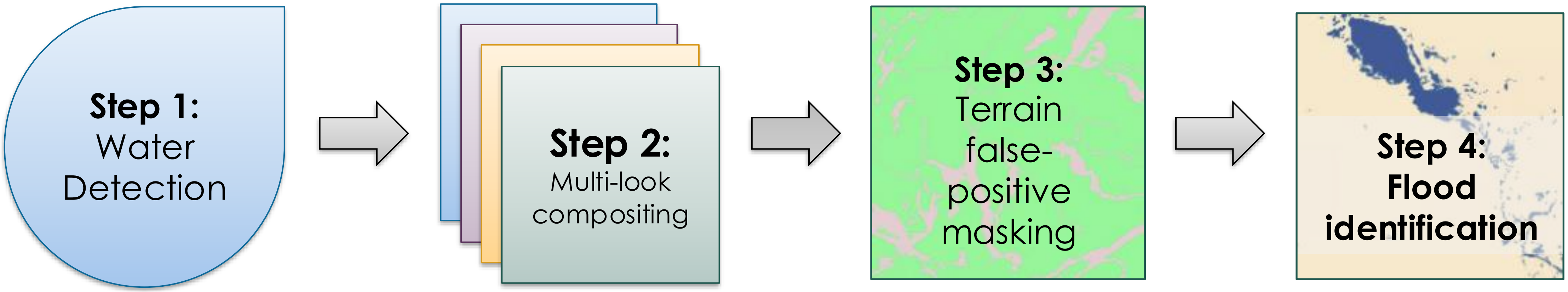
MODIS/Aqua
February, 1:30 PM

2. General topographic mask
   HAND: Height Above Nearest Drainage



HAND identifies areas unlikely to retain flood water, given 250 m resolution pixels (green = masked area)

Flood Identification



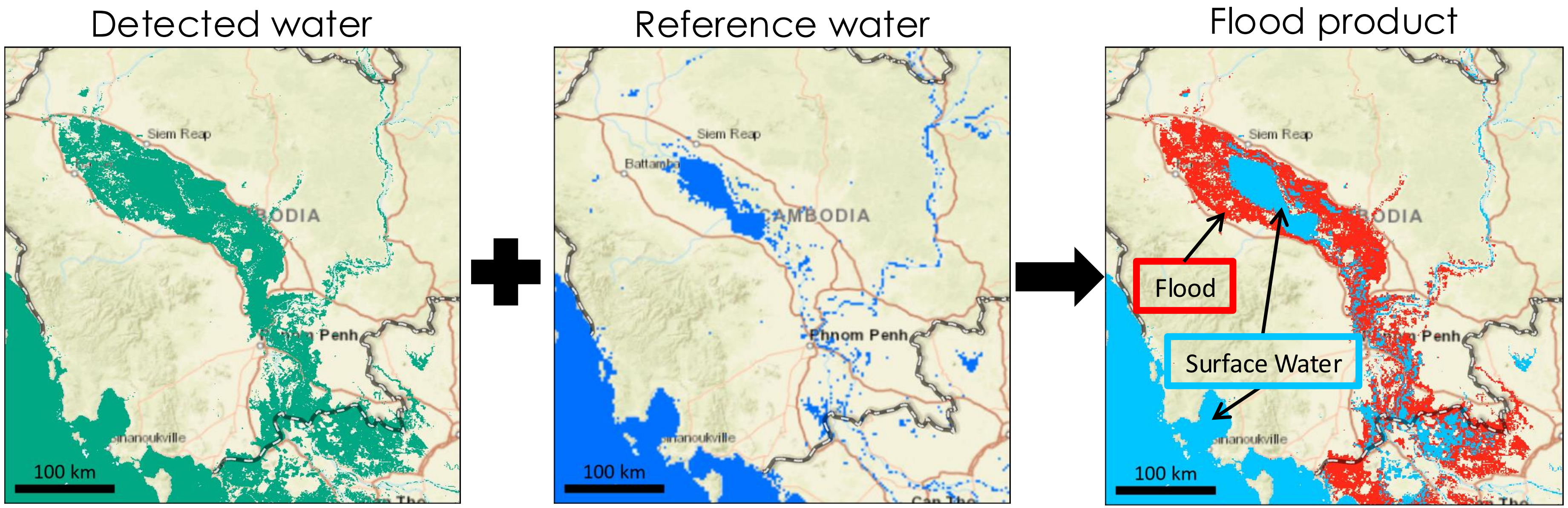
Flood identification

1. Compare detected water to reference water layer (=expected water).

   Reference water provided by yearly MOD44W land-water product.

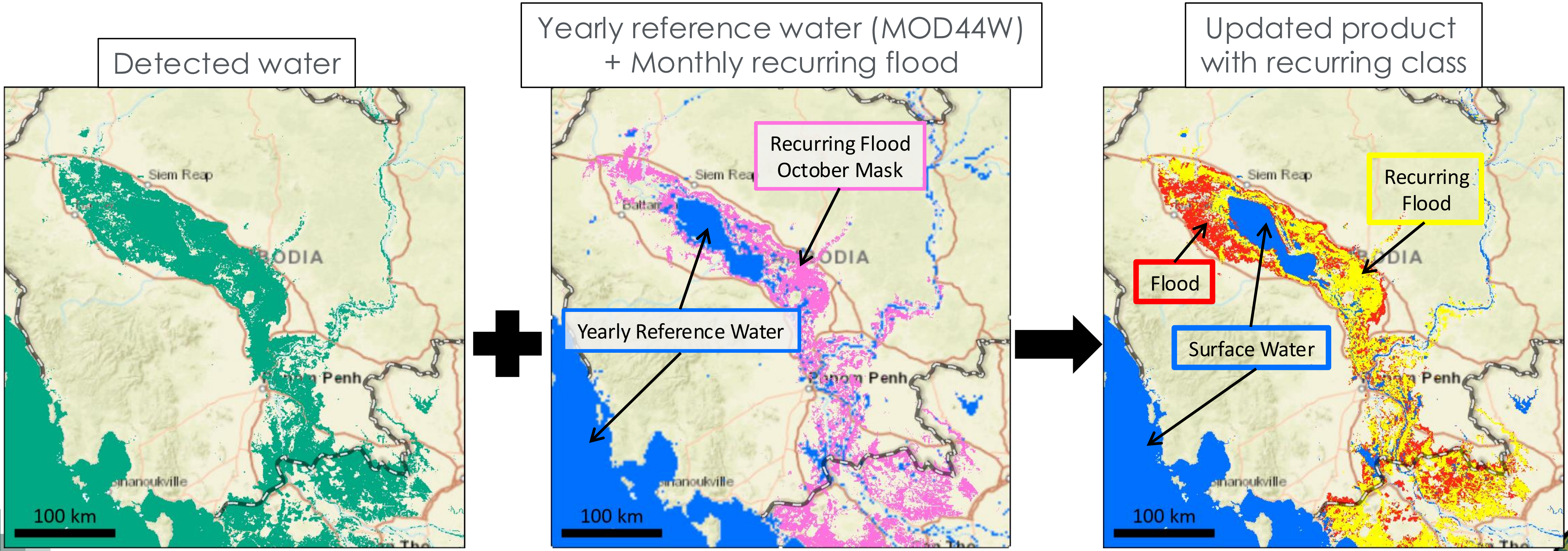
2. Water not matching reference: ➔FLOOD

3. Water matching reference: ➔SURFACE WATER



+ Identification of “Recurring Flood”

- New feature, introduced with Release 1.1 in Dec 2025
- Monthly recurring flood masks based on 22-year product history (2003-2024)
- Helps discriminate regularly occurring seasonal flood from unusual flooding

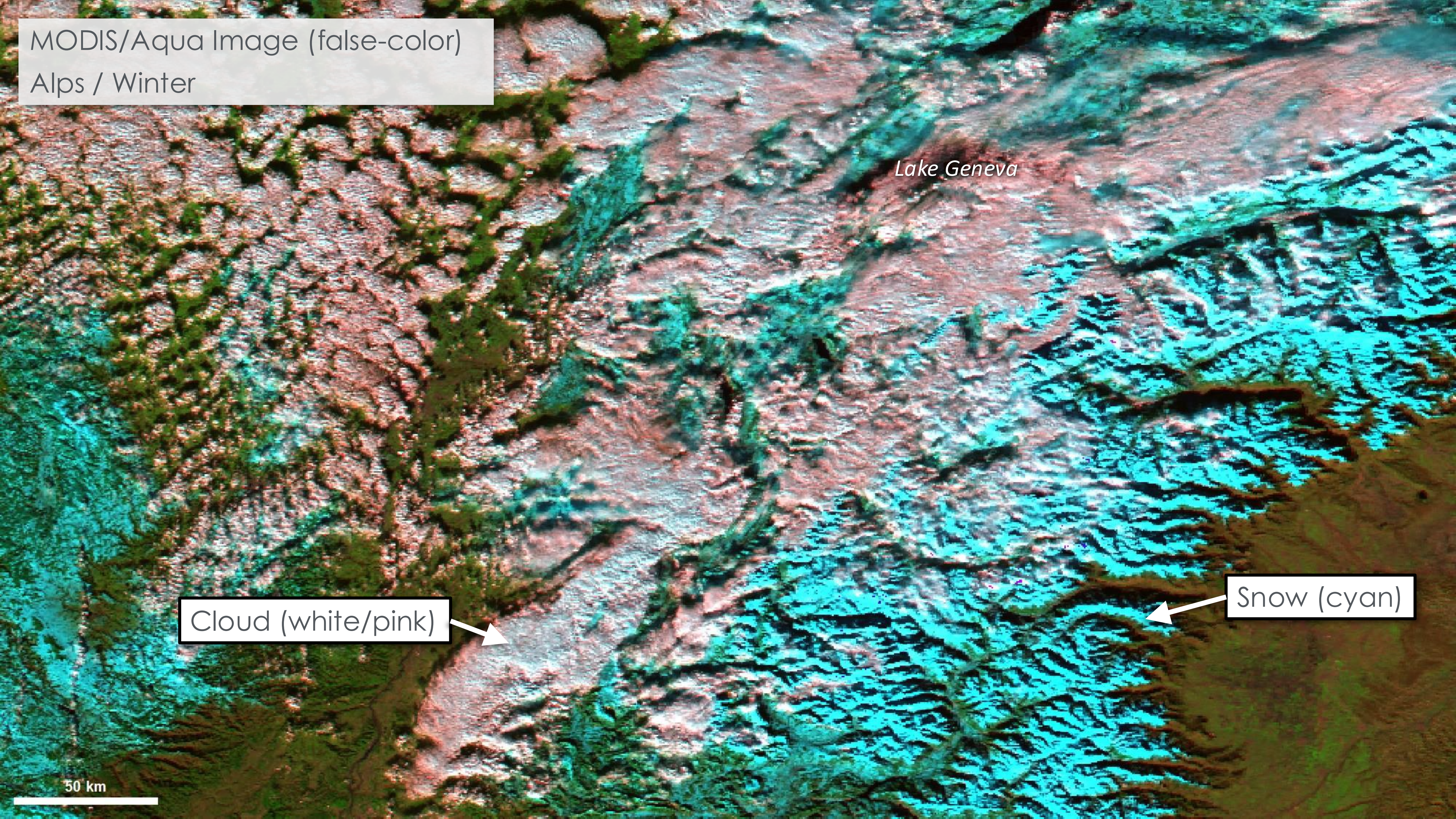


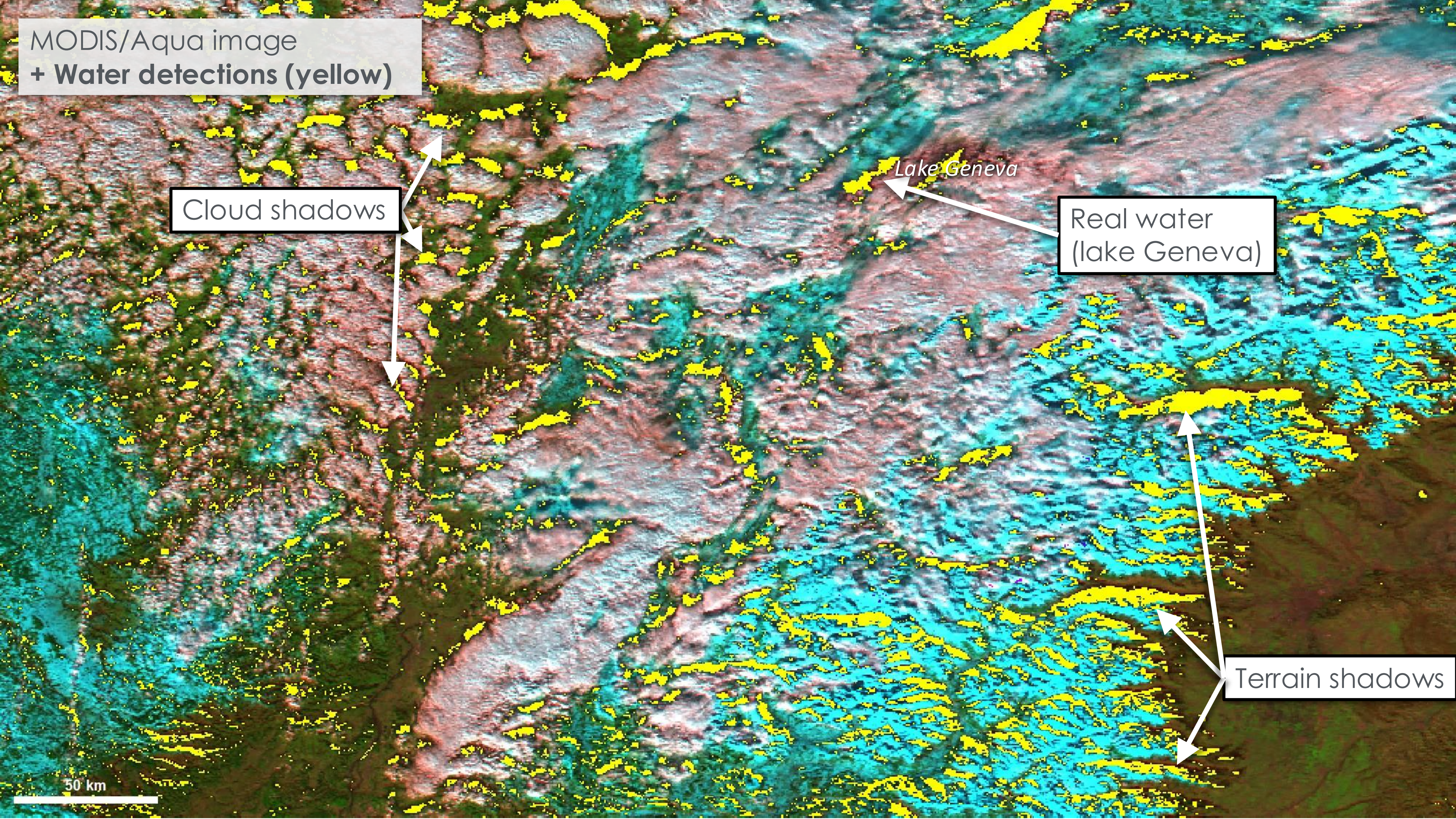
Flood Product Approach: Details

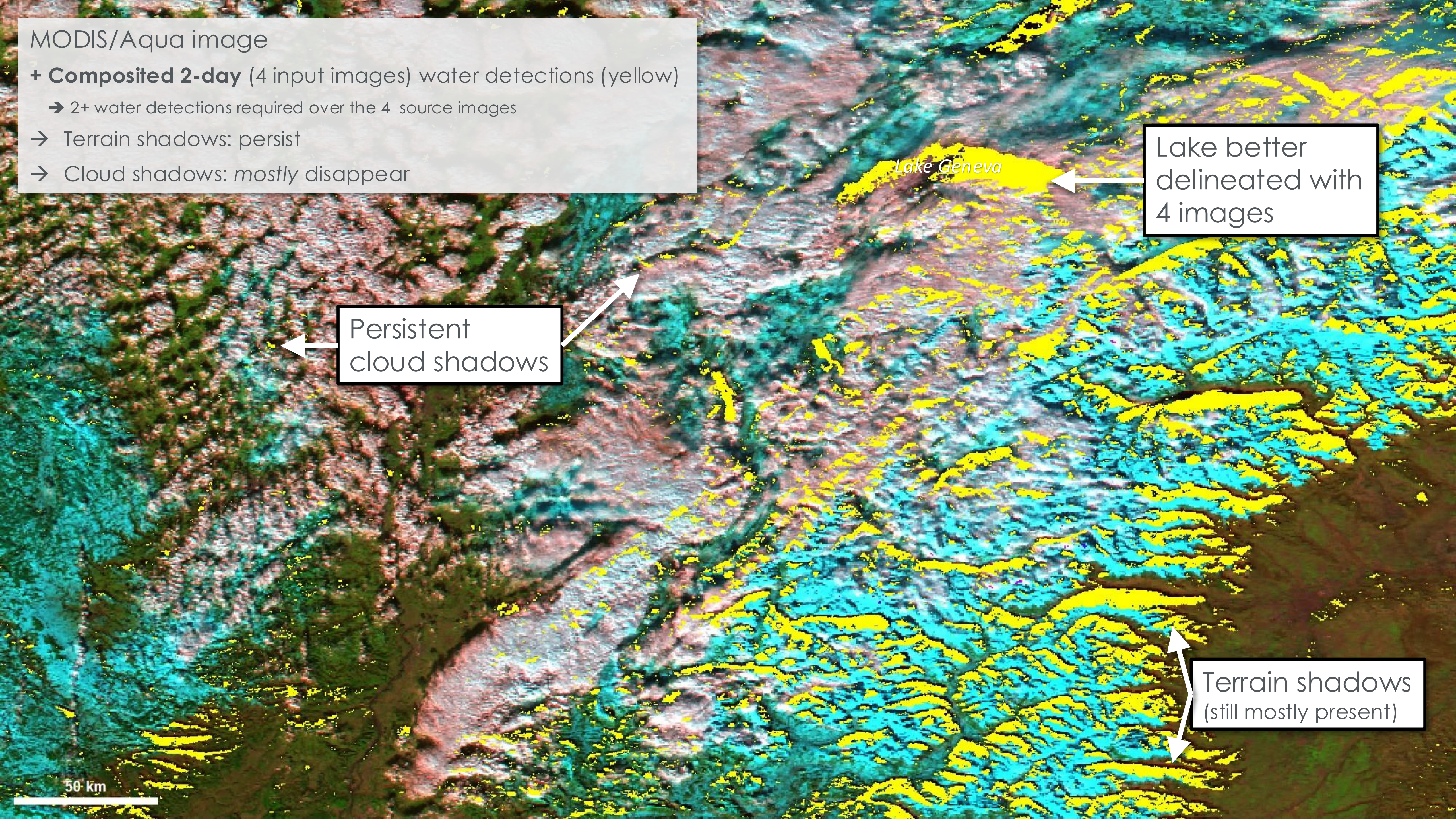
Detail: False-positive Masking

- Shadows look like water → false positives!
  - Shadow is very difficult to discriminate from real water in red and near-infrared wavelengths
- Terrain shadows
  - In mountainous areas, mostly in winter
  - First cut: apply computed terrain shadow masks (useful but not perfect)
  - Second cut: apply HAND topographic mask (masks areas where water should not be able to accumulate due to nearby drainage potential)
- Cloud shadows
  - Multi-look compositing eliminates most, but the 1-day product (which usually will not require > 1 water observation to mark a pixel as water) can be heavily contaminated
  - However: shadows will occasionally recur in the same location, leading to false-positives in 2 and (much less frequently) 3-day products

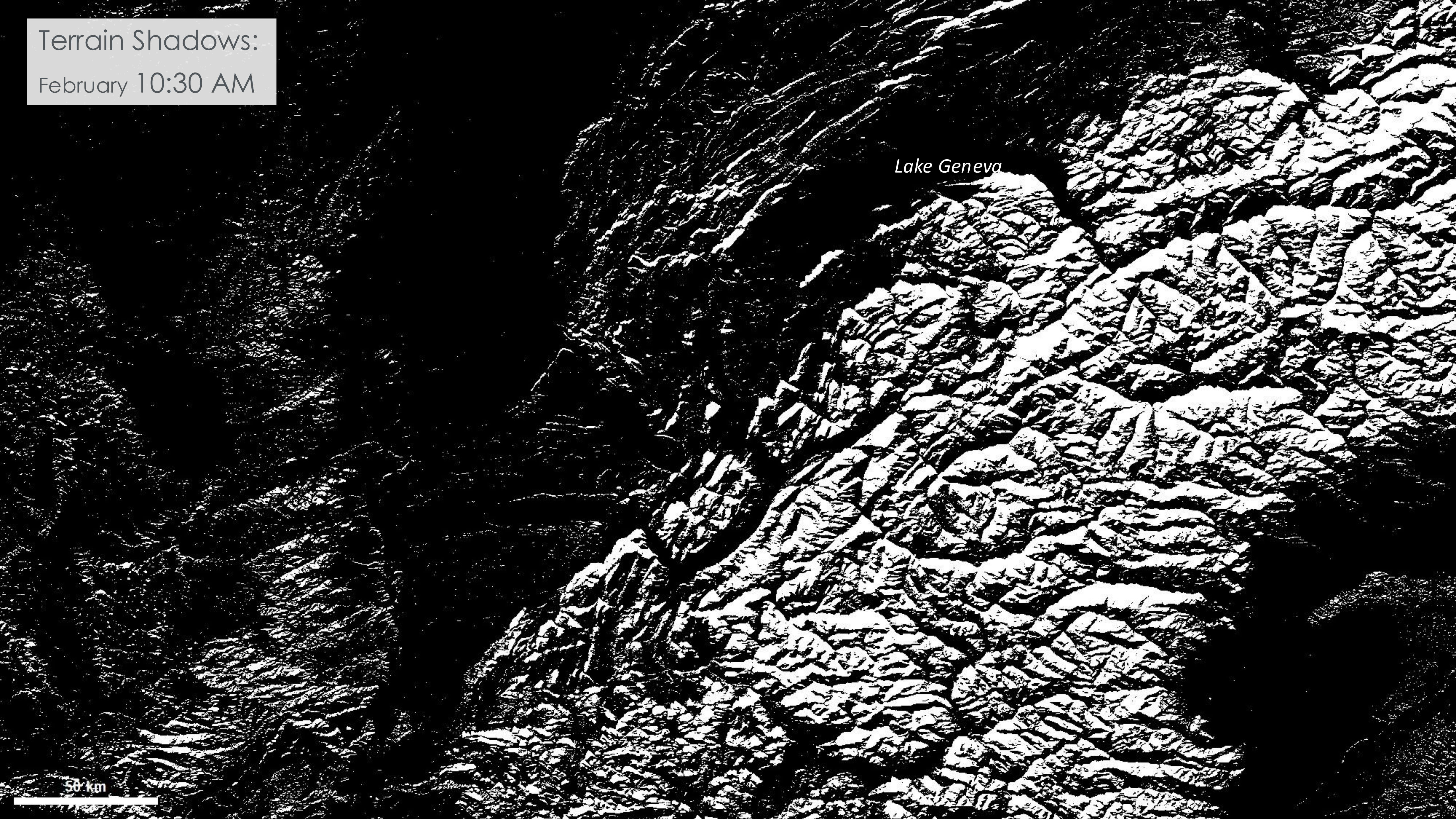
➔ **Reviewing source imagery is critical to understanding potential cloud contamination!**

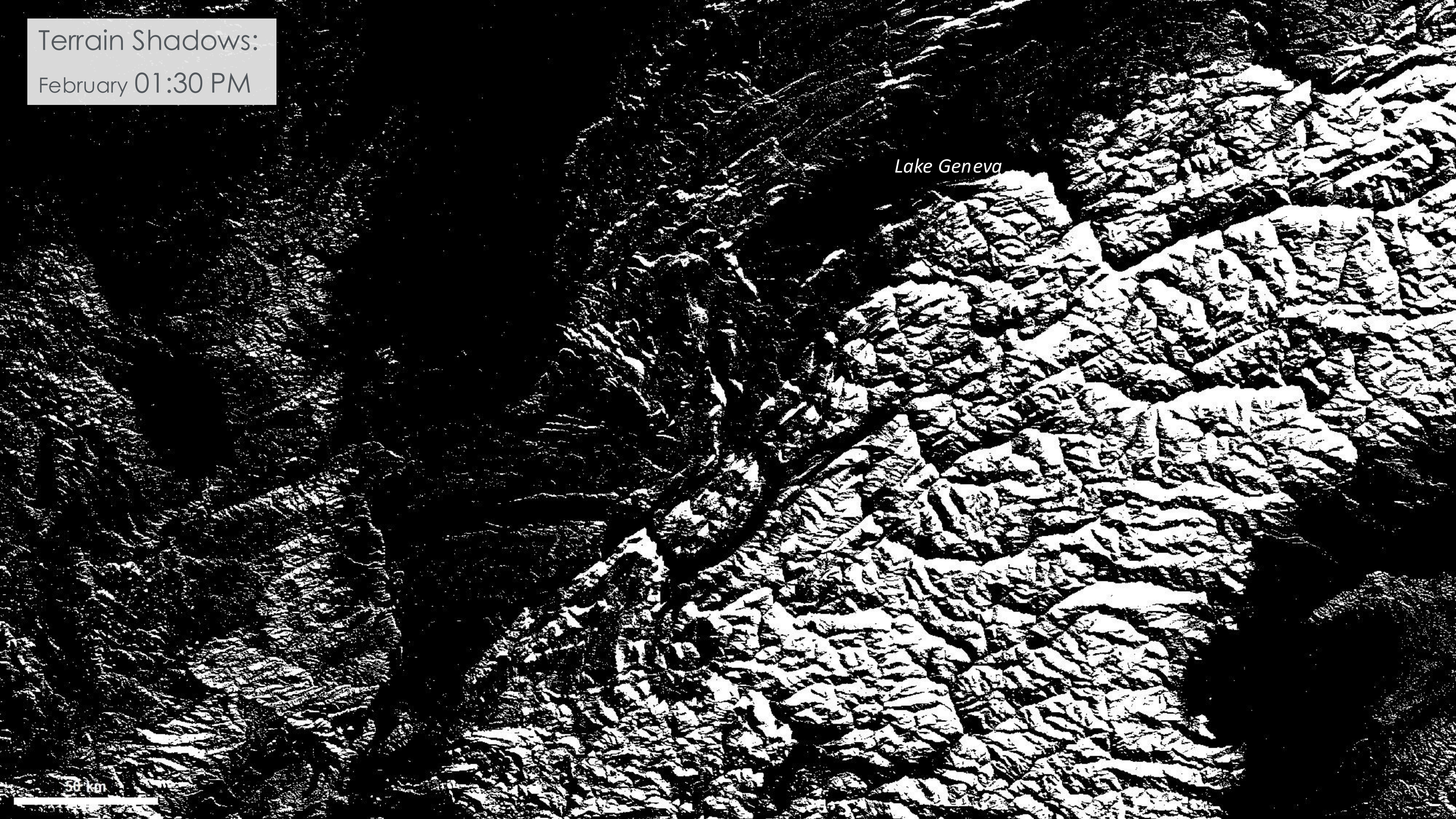

MODIS/Aqua Image (false-color)
Alps / Winter
Lake Geneva
Cloud (white/pink)
Snow (cyan)
50 km

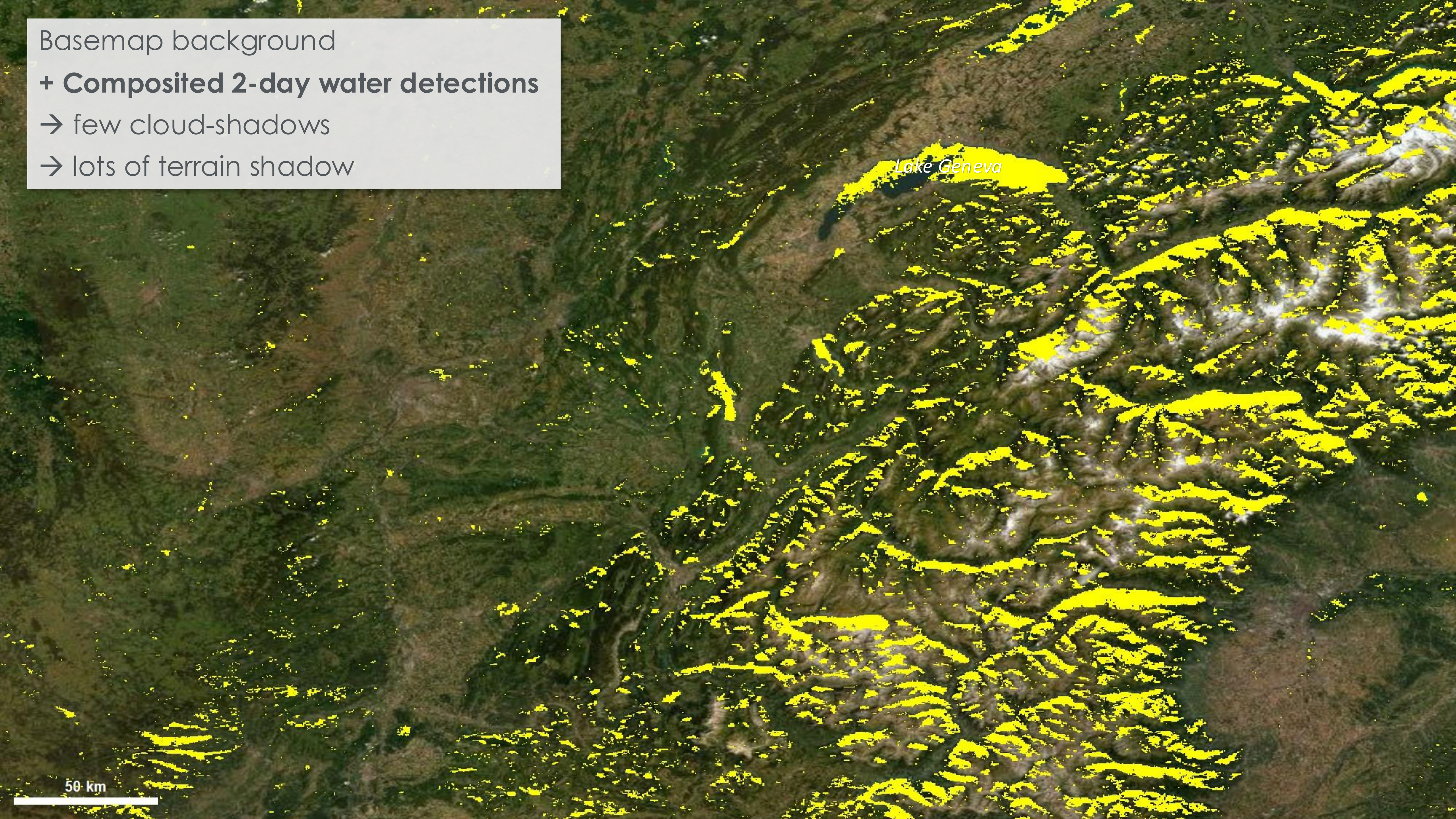

MODIS/Aqua image
+ Water detections (yellow)
Cloud shadows
Lake Geneva
Real water
(lake Geneva)
Terrain shadows
50 km

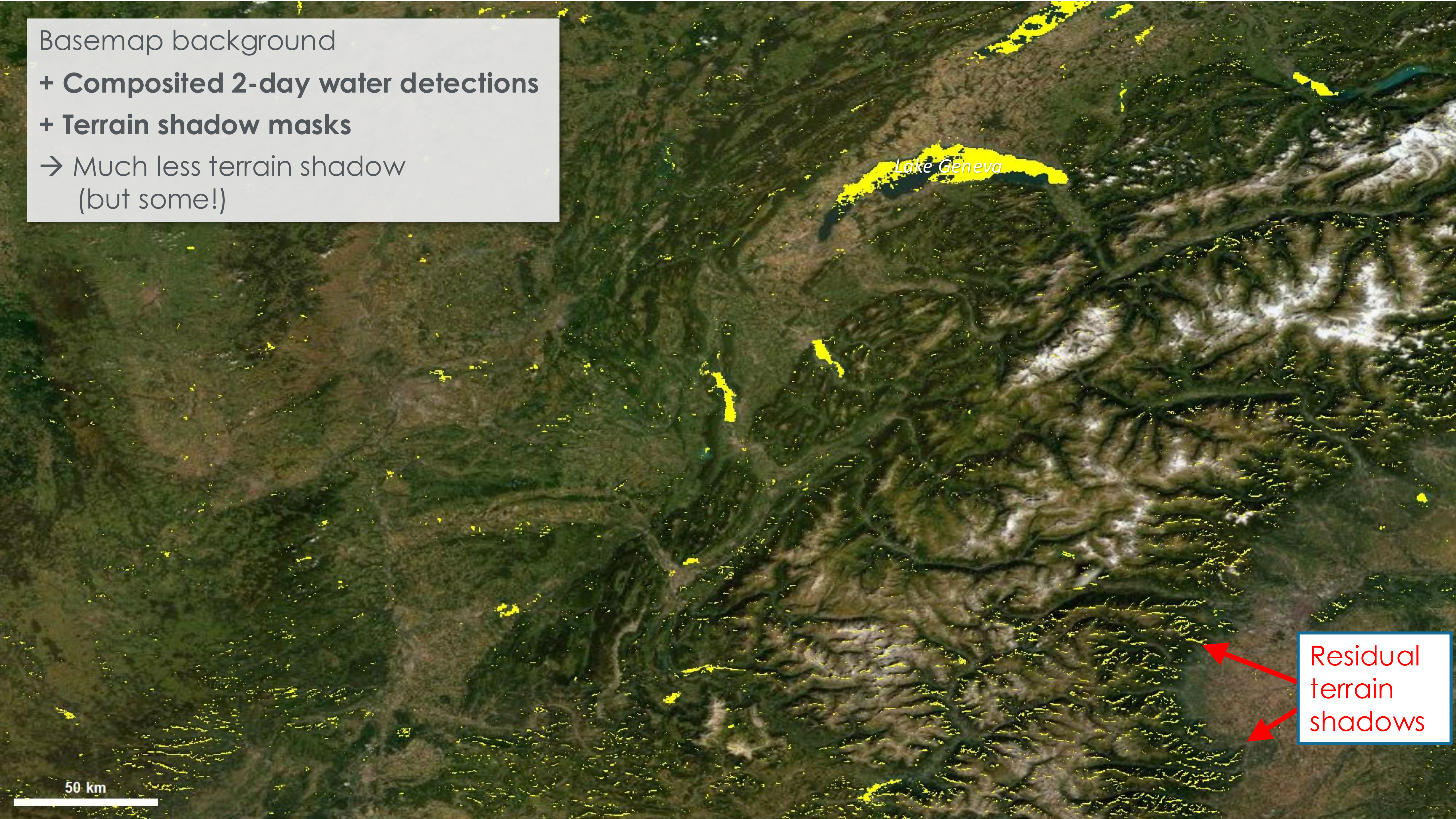

MODIS/Aqua image
+ Composited 2-day (4 input images) water detections (yellow)
→ 2+ water detections required over the 4 source images
→ Terrain shadows: persist
→ Cloud shadows: mostly disappear
Lake Geneva
Lake better delineated with 4 images
Persistent cloud shadows
Terrain shadows
(still mostly present)
50 km

Example: Alps - terrain


Terrain Shadows:
February 10:30 AM
Lake Geneva
50 km

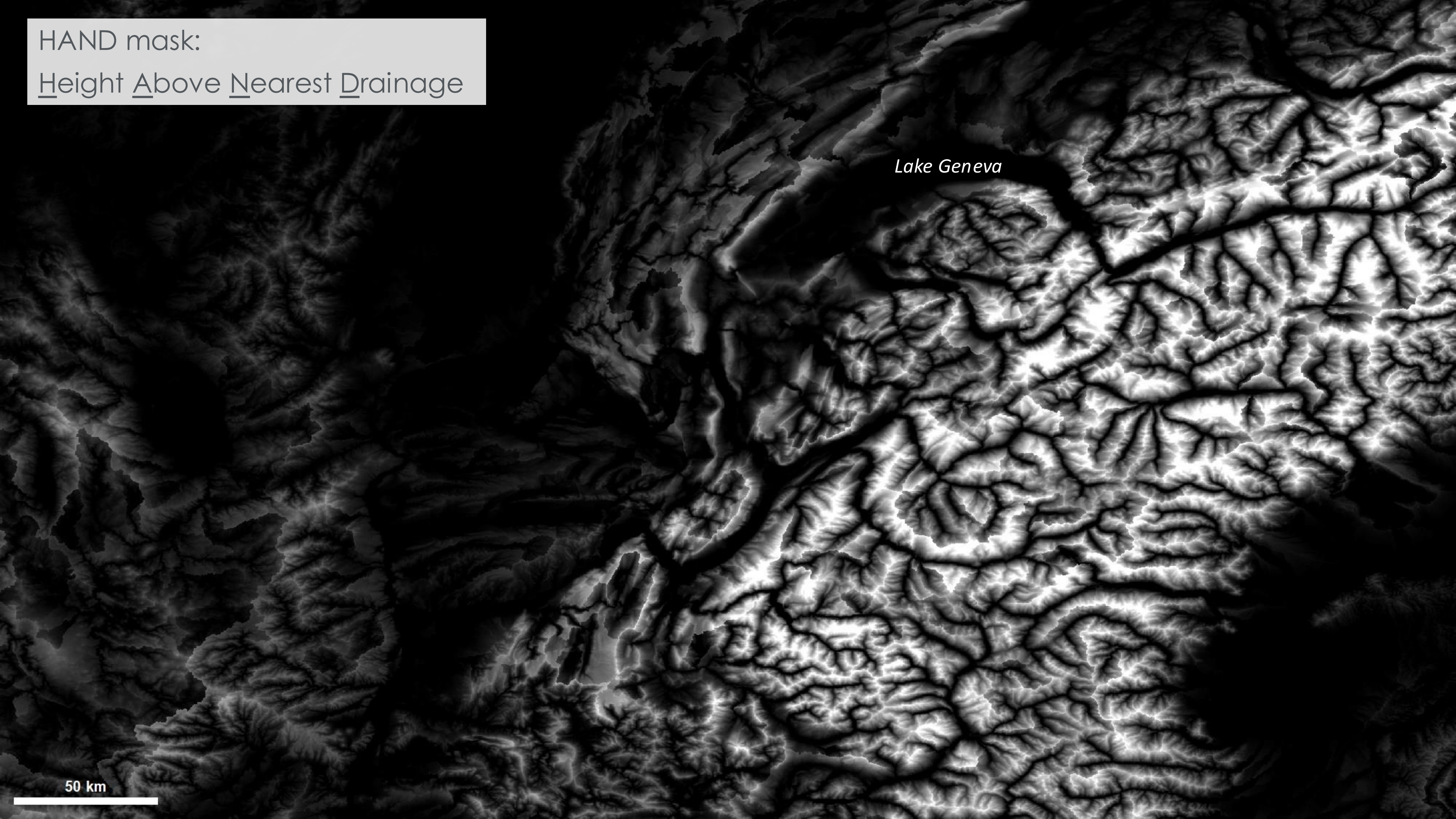

Terrain Shadows:
February 01:30 PM
Lake Geneva
50 km

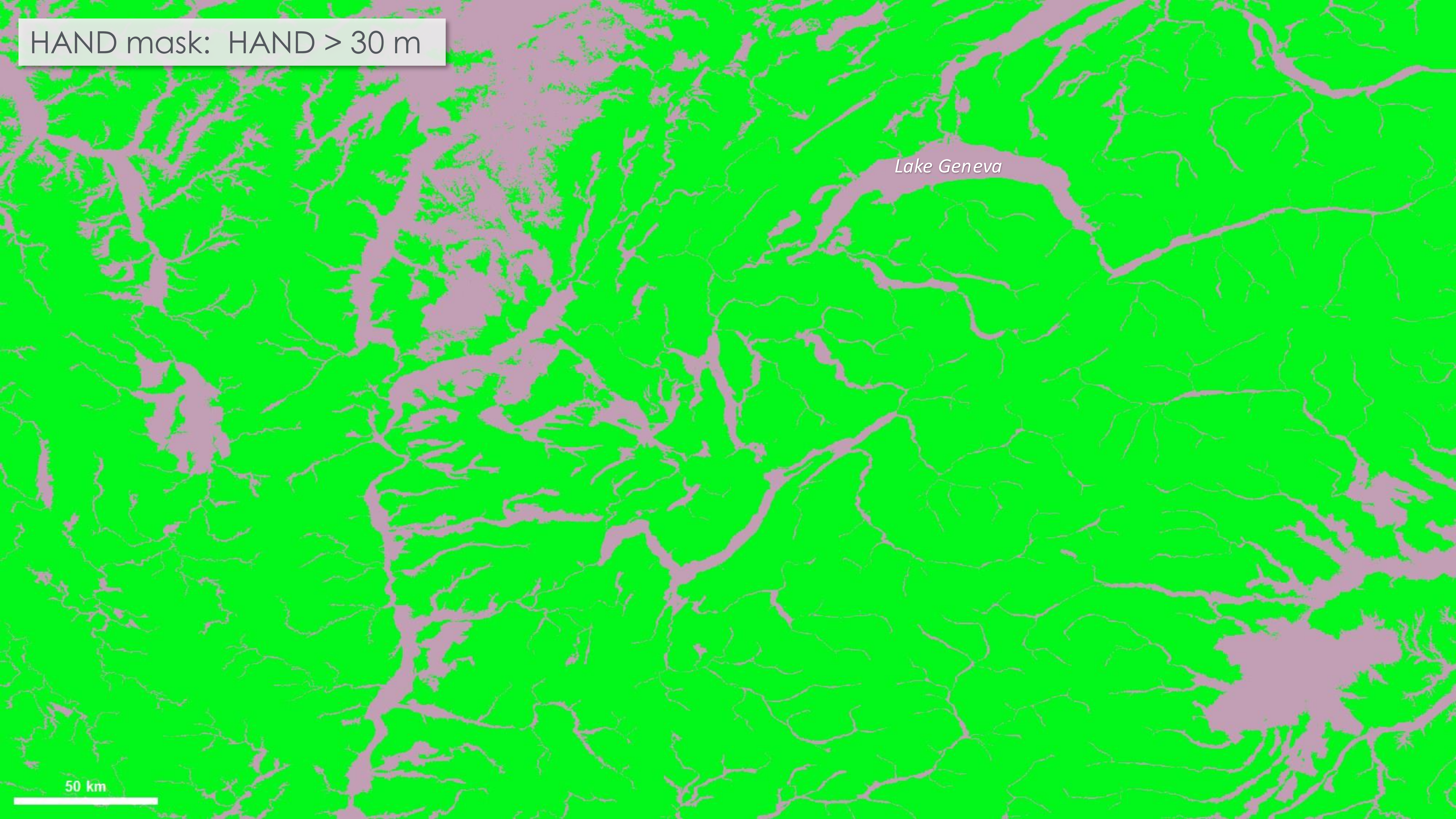

Basemap background
+ Composited 2-day water detections
→ few cloud-shadows
→ lots of terrain shadow
Lake Geneva
50 km


Basemap background
+ Composited 2-day water detections
+ Terrain shadow masks
→ Much less terrain shadow
(but some!)
Lake Geneva
Residual terrain shadows
50 km

HAND mask:

Height Above Nearest Drainage



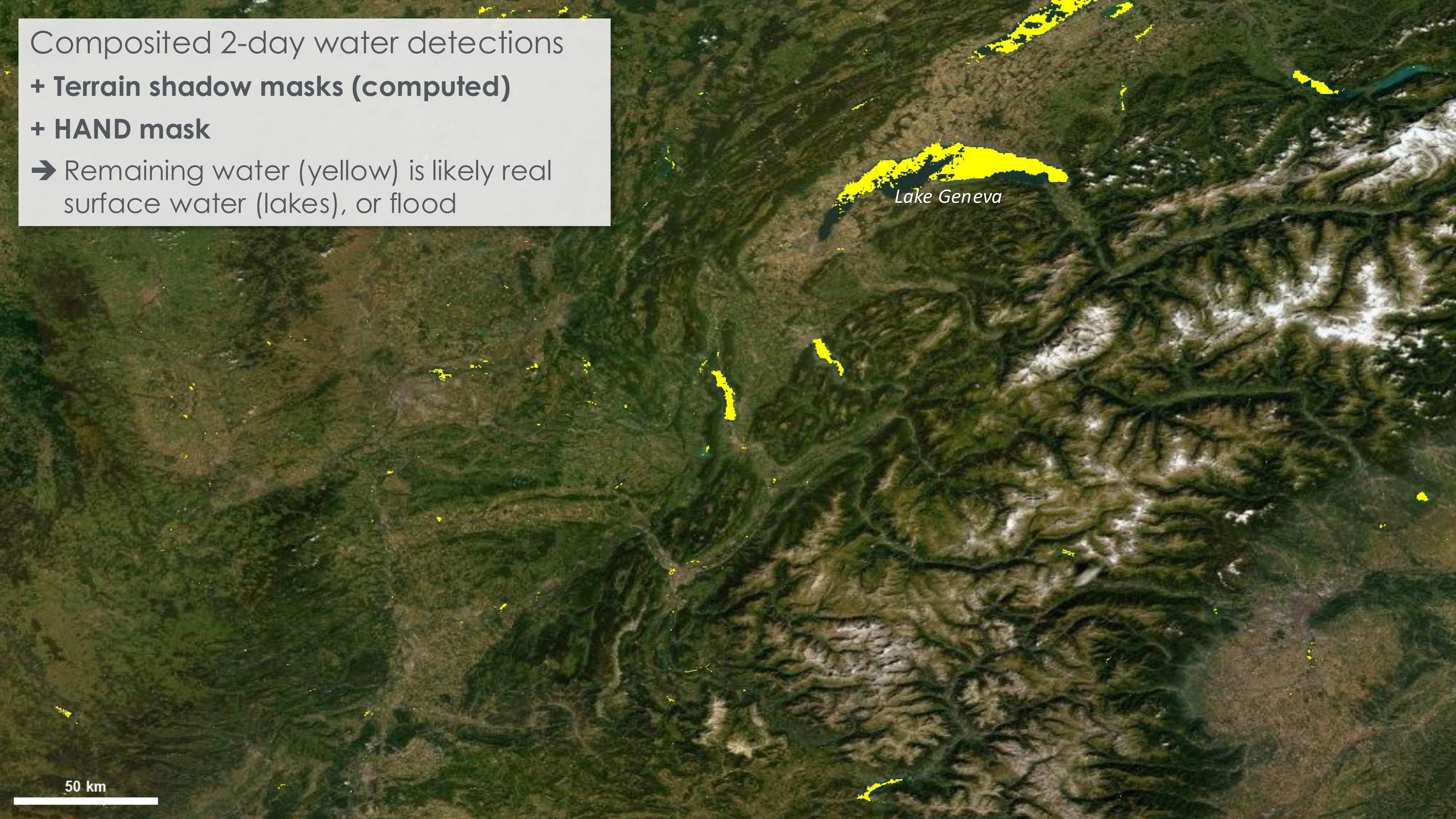

HAND mask: HAND > 30 m
Lake Geneva
50 km

Composited 2-day water detections

+ Terrain shadow masks (computed)

+ HAND mask

➔ Remaining water (yellow) is likely real surface water (lakes), or flood



Detail: Multi-look Compositing

- **How it works:**
  - Sum water detections from all observations, over 3 different composite periods: 1 day, 2 days, and 3 days.
  - If SUM >= THRESHOLD: mark pixel as water
    - The threshold is generally equal to half the number of observations:
      - 1 for 1-day product (typically 2 observations)
      - 2 for 2-day product (typically 4 obs)
      - 3 for 3-day product (typically 6 obs)

- **1-day product:** requires JUST 1 water detection from either of the 2 observations available over 1 day.
  - Most up-to-date but **WILL probably contain cloud-shadow false-positives IF cloud is present.**
- **2-day product**: requires 2+ water observations over 2 days of imagery.
- **3-day product**: requires 3+ observations over 3 days. Minimal false-positives. Potentially less up-to-date.

Which Composite Flood Products to Use?

Which Product to Use?

- Depends on cloud conditions, and:
  - Tolerance for false positives and false negatives
  - Need for only the most up-to-date information

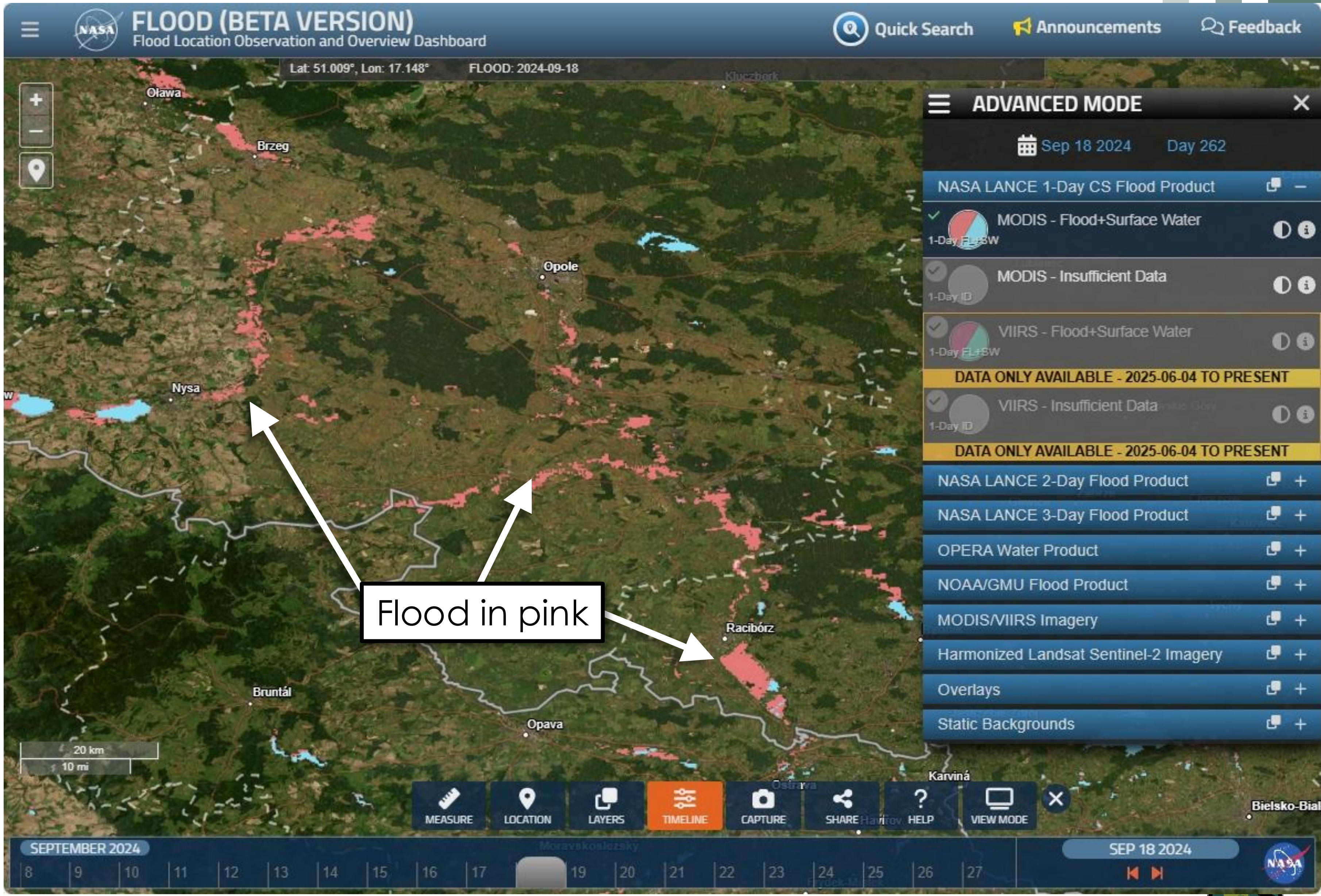
- Generally, a good balance between cloudiness and timeliness: **2-day**
- Clear conditions? Verify visually in Worldview → **Use 1-day**
- Very sensitive to false-positives and/or timeliness is not critical? → **Use 3-day**
- Need the latest info? → **Use 1-day**, **but CHECK CLOUDS!**
- **Best approach? Review visible imagery for date/area of interest in conjunction with the products**

Composite Use Example: 18 Sept 2024 Southern Poland

1-Day Composite

Basemap background

- Lots of flood
- Appears to follow linear-ish features (suggesting rivers): ✓

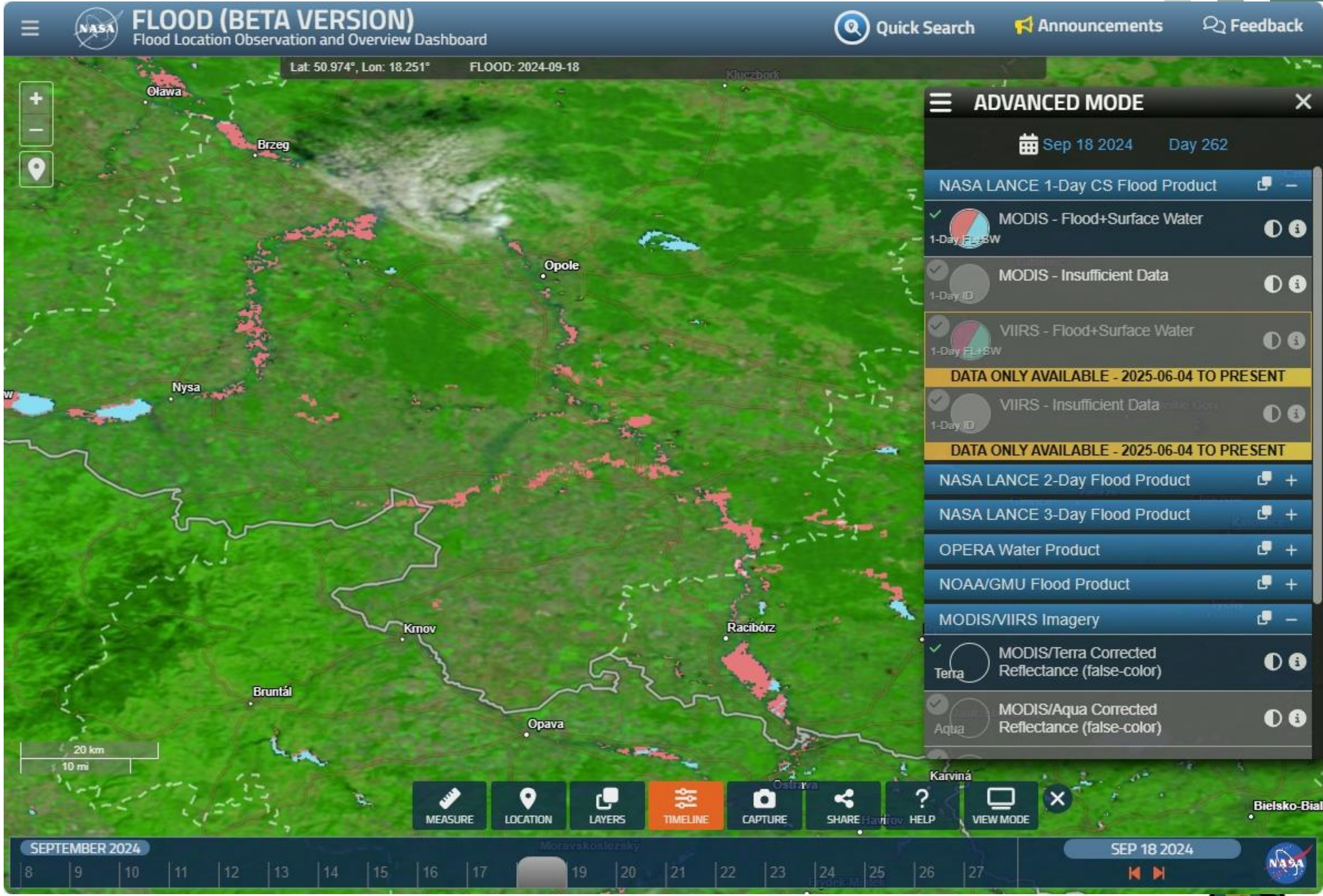


Composite Use Example: 18 Sept 2024 Southern Poland

1-Day Composite

MODIS/Terra image background

- Still looks reasonable.
- Minimal cloud; no obvious cloud issues: ✓



Composite Use Example: 18 Sept 2024 Southern Poland

1-Day Composite

Change to MODIS/Aqua image

- "Flood" reported directly aligned with cloud shadow !!
- Some flood does appear reasonable

Composite Use Example: 18 Sept 2024 Southern Poland

2-Day Composite

- MODIS/Aqua image background
- Most cloud shadow false-positives have disappeared

For 2-day composite, 4 images should be reviewed:

- Terra: Current day
- Aqua: Current day
- Terra: Previous day
- Aqua: Previous day

Similarly for VIIRS: review NOAA-20 and NOAA-21 in place of Terra and Aqua

Composite Use Example: 18 Sept 2024 Southern Poland

3-Day Composite

MODIS/Terra image background

- All cloud shadow false-positives have disappeared!
- But much real flood has also disappeared!

→ 2 days ago, area was entirely cloudy, so few pixels meet 3-day water-detection threshold

16 Sept 2024 Terra: FULL CLOUD

Recent Updates

Recurring Flood

Objective: Distinguish recurring flood from anomalous events

Motivation: Standard annual water masks fail to account for seasonal or ephemeral flooding

Data source: reprocessed 22-year MODIS flood product archive, 2003-2024

Temporal framework: Monthly analysis using rolling 3-month windows (target month +/-1) to capture seasonal hydrology

Statistical filtering: Filters out spurious detections caused by sparse data or sampling noise

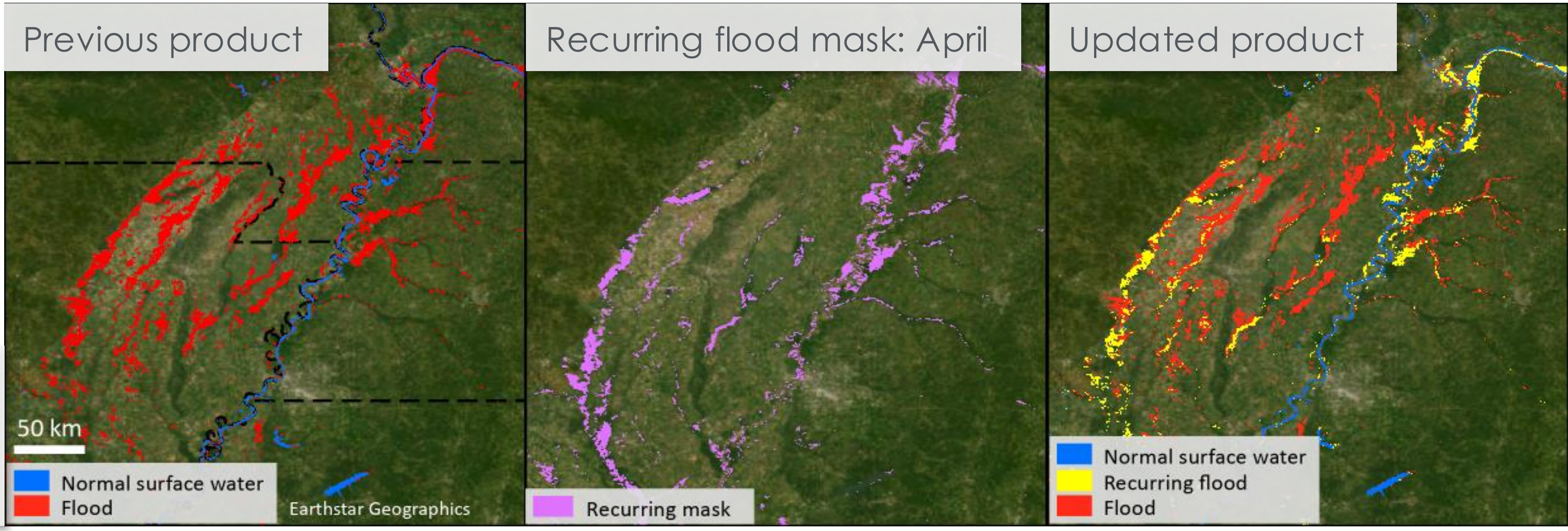
Classification criteria:

1. **Per year:** Require >= 3 flooded days within the 90-day window.
2. **Per pixel:** Require yearly criterion to be met in >= 7 of the 22 years. (~1/3 of the data record)

Post-processing: 3×3 majority filter applied to minimize noise and fill small spatial gaps

Implemented with product release 1.1 (12 Dec 2025), and applied to reprocessed historical archive

Mississippi river / Midwest flooding
8 April 2025
VIIRS 1-day product

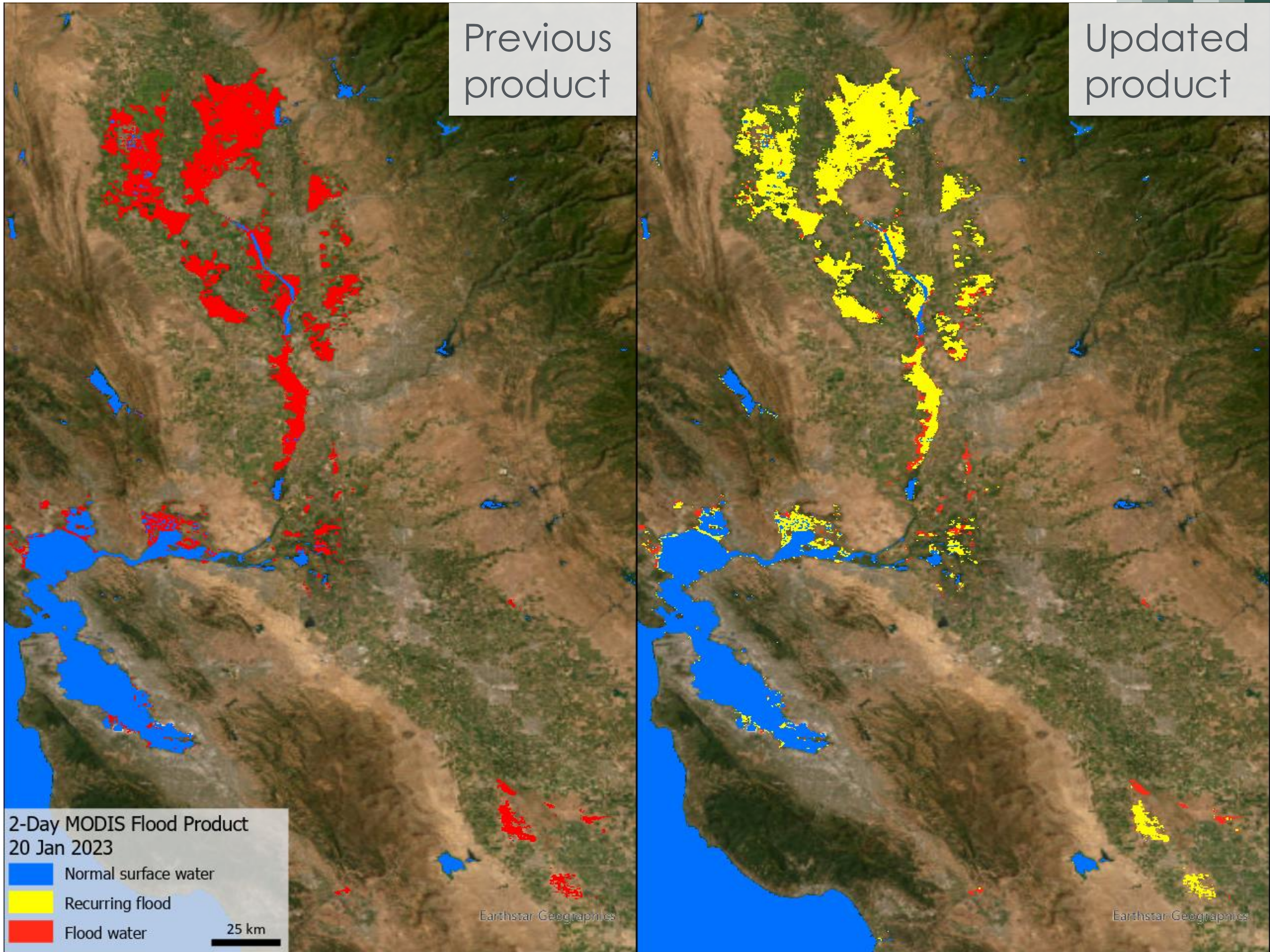


Recurring Flood Example

N California example January 2023

Recurring flood (yellow): water was detected in this 20 Jan 2023 MODIS product, and has also been detected here within the Dec-Jan-Feb period, in 7 or more years from the 2003-2024 archive (eg, >= 1/3 of archive)

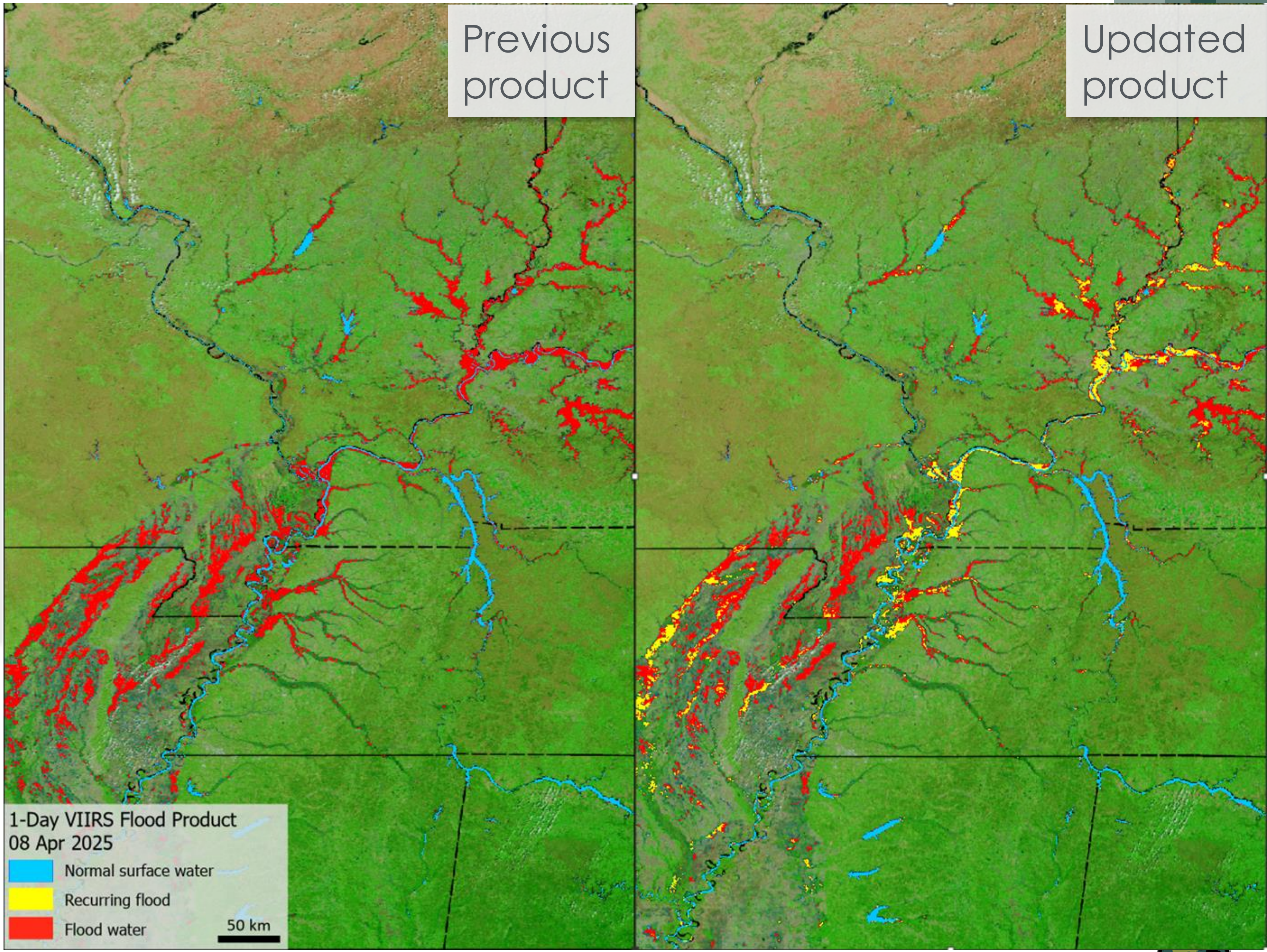
→ Although substantial areas of flood were detected on this date, most of this is regularly recurring flood.



Recurring Flood Example

Mississippi – Ohio example April 2025

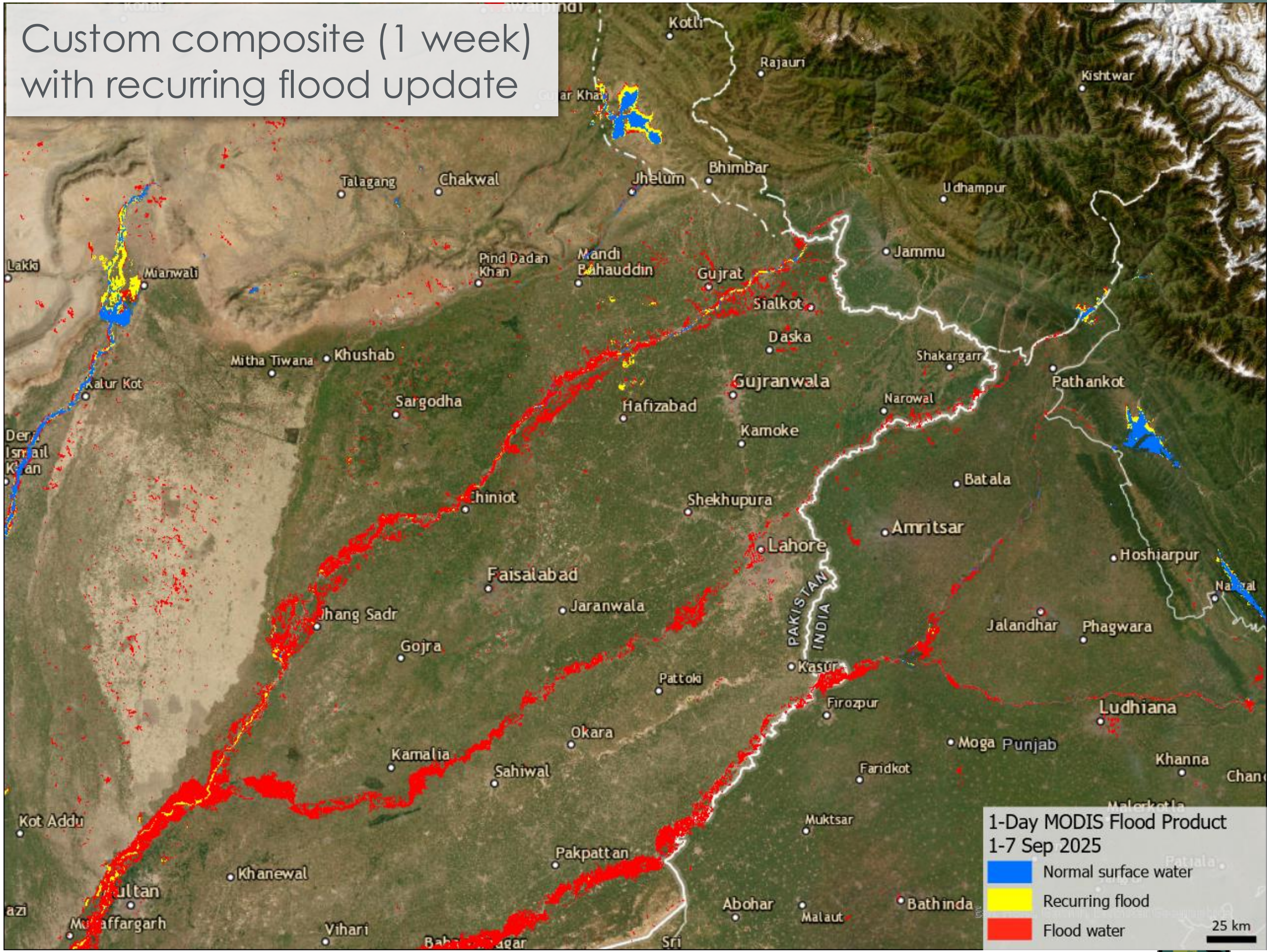
→ Substantial areas of flood were detected, most of which was **NOT recurring flood**, but reflects unusual flooding.



Recurring Flood Example

Indus/Punjab river system Sept 2025

→ Mostly ***not*** recurring, but unusual flooding in this event



Evaluation

Validation/Evaluation

Purpose:

- Is water detection algorithm correctly detecting water that is visibly obvious?
- Are certain events or situations problematic?
- Do we see differences between detection of flood vs of non-flood water?

Caveat:

- No rigorous ground-truth dataset: ground truth is difficult to find, expensive to collect, and biased towards accessible locations

Method:

- Manual qualitative assessment, using MODIS or Landsat visible imagery to inform.
- 50 events selected from DFO master list of floods.
    - Global distribution.
    - Including areas with high and low cloud cover (humid tropics to arid regions).
    - Varying landcovers.
- + 50 permanent water sites for evaluation of surface water detection, apart from flood.
- Conducted for legacy product in 2014.

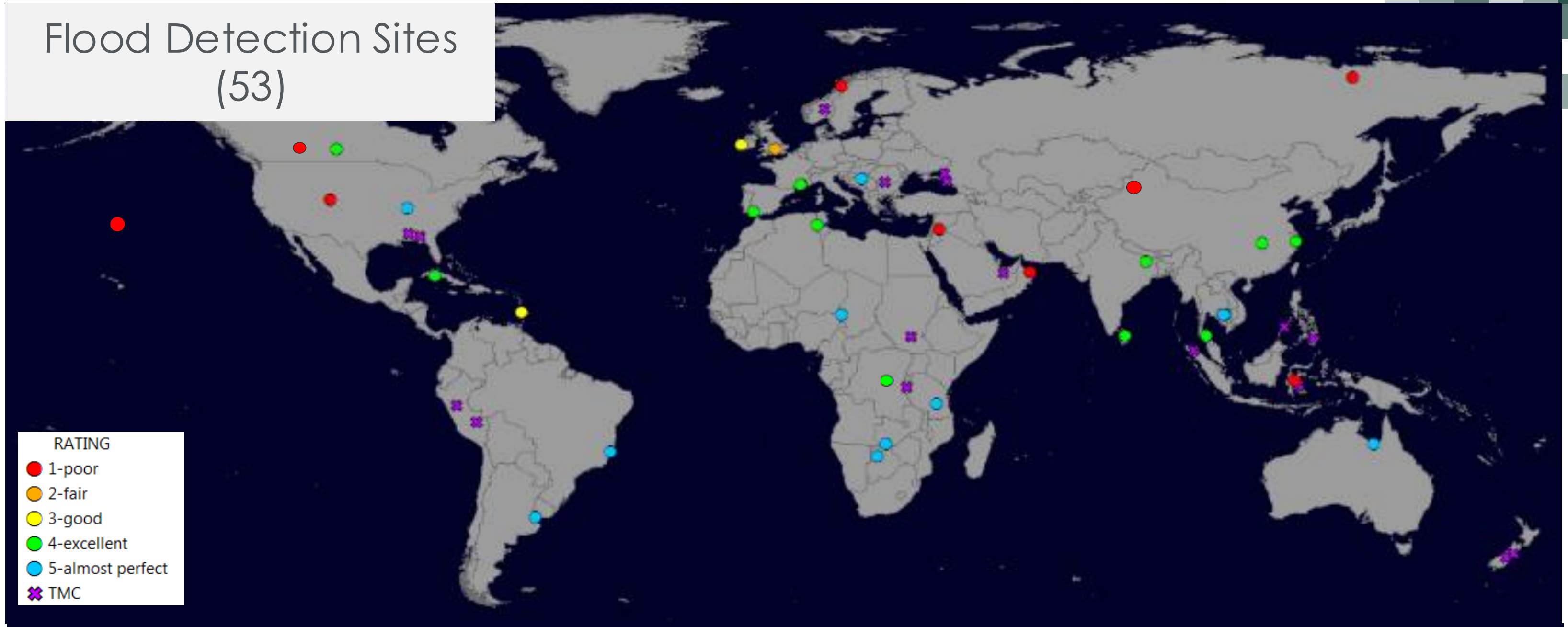
Evaluation Results

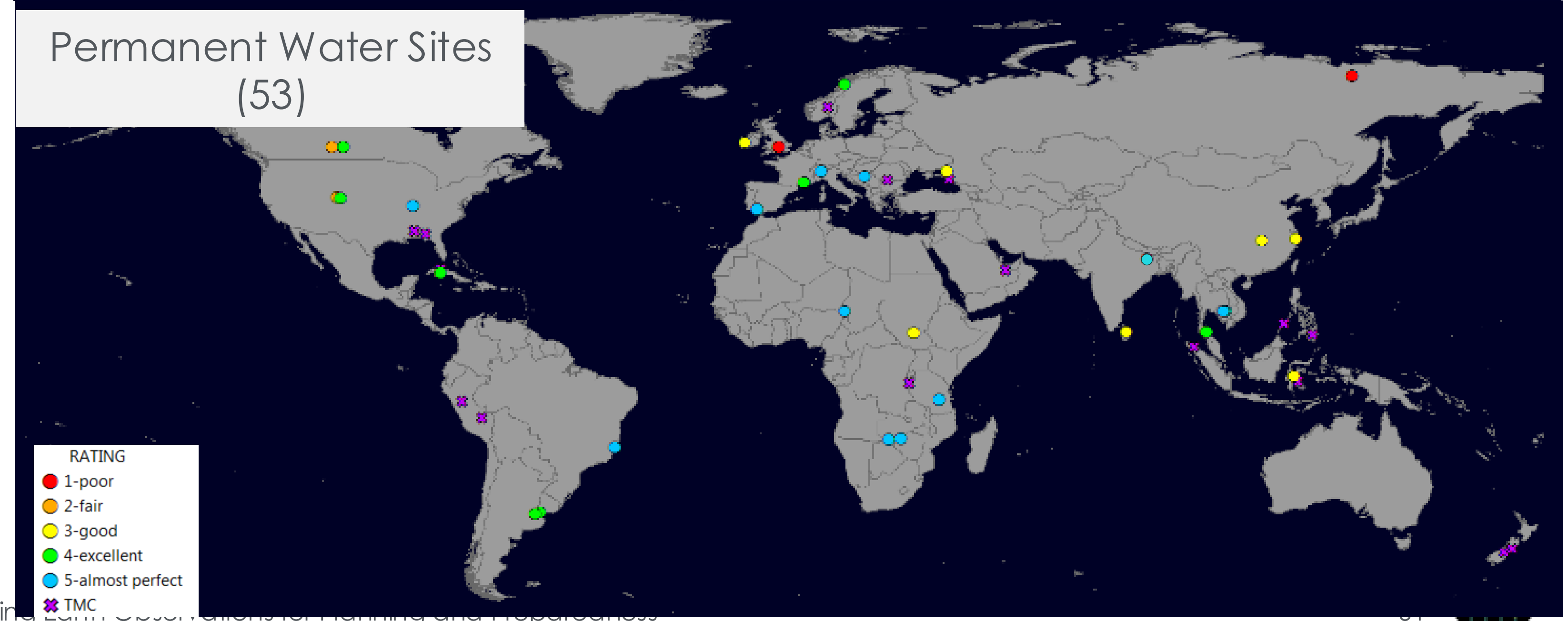
Flood Ratings

RATING	Count	%
Good or better	**23**	**66**
Fair or poor	12	34
Total Assessable	35	100
Too cloudy	17	33
TOTAL EVENTS	53	

Permanent Water Ratings

RATING	Count	% Clear
Good or better	**31**	**85**
Fair or poor	6	16
Total Assessable	37	100
Too cloudy	16	30
TOTAL EVENTS	53	

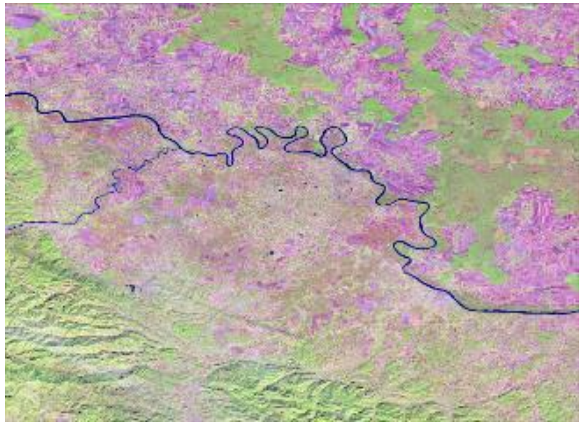




Correct Water & Flood ID

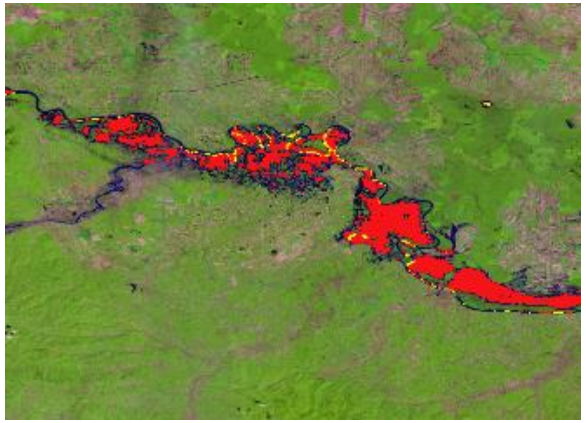
Bosnia and Herzegovina: 23 May 2014

Base map



Landsat 8 Pre-flood
October 19, 2013

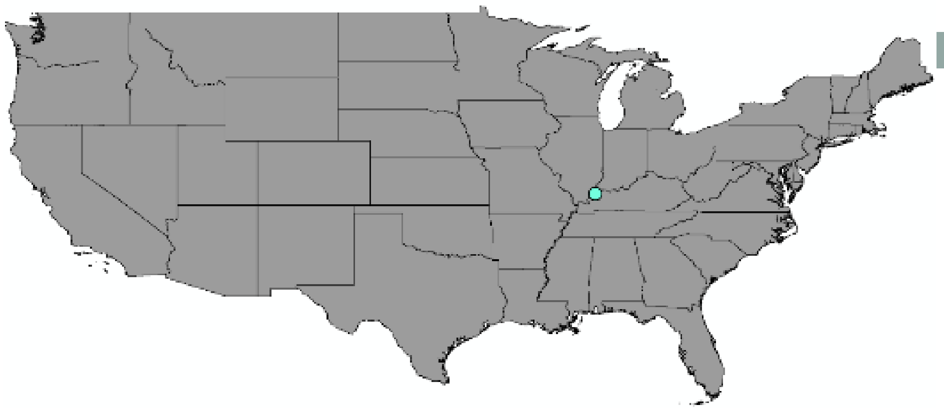
Landsat 8 Flood
May 22, 2014



MODIS NRT product
May 22, 2014

Flood water
Non-flood water
Insufficient Data

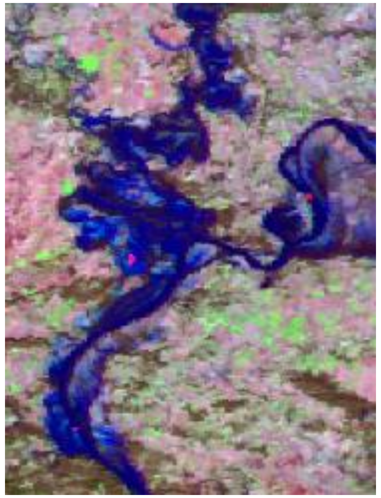
Correct Water & Flood ID



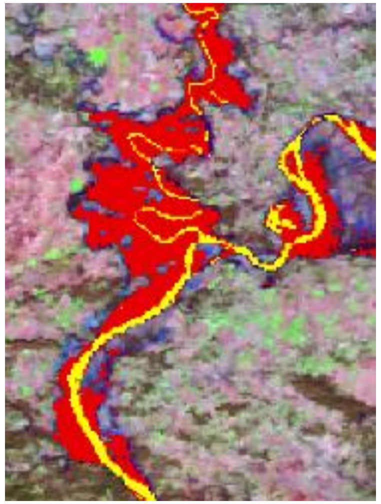
Kentucky: 04 Jan 2014

Base map

MODIS Pre-Flood
Oct 12, 2013



MODIS Flood
Jan 4, 2014



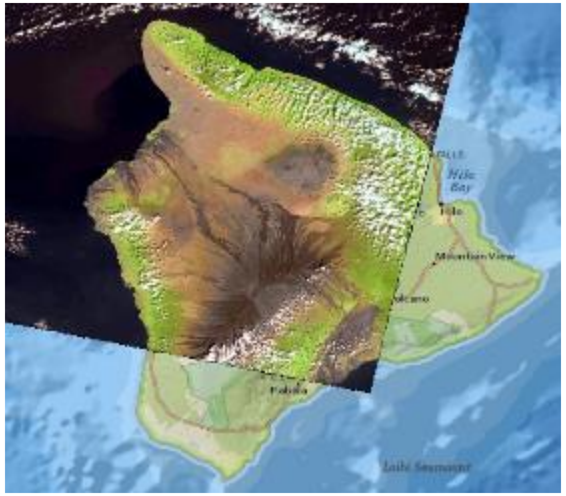
MODIS NRT Product
Jan 4, 2014

- Flood water
- Non-flood water
- Insufficient Data

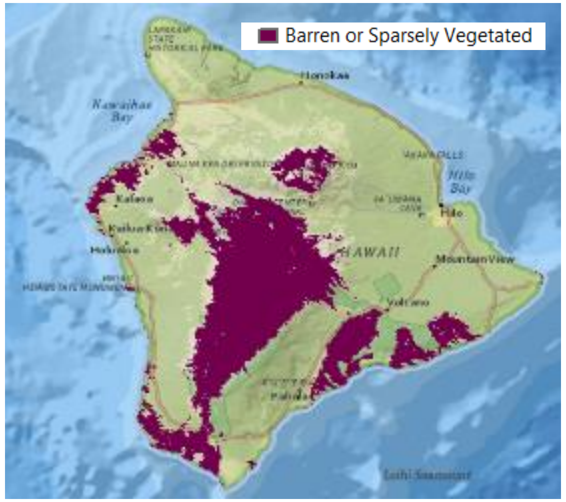
False-Positive: Dark Volcanic Rock

- Volcanic masks for a few locations.
- For Mauna Loa, HAND mask largely solved this problem.

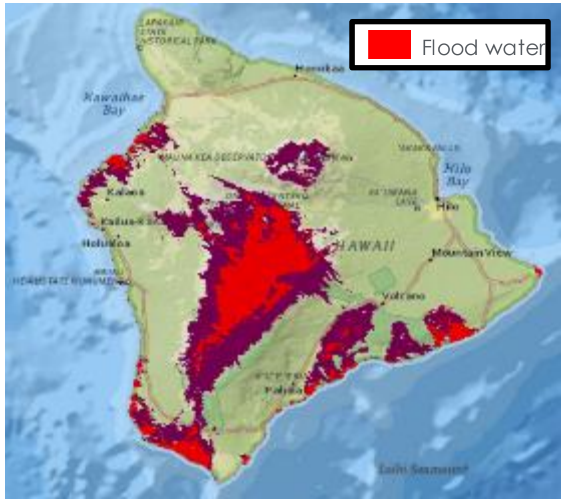
Mauna Loa, **Hawaii:** 17 Dec 2013



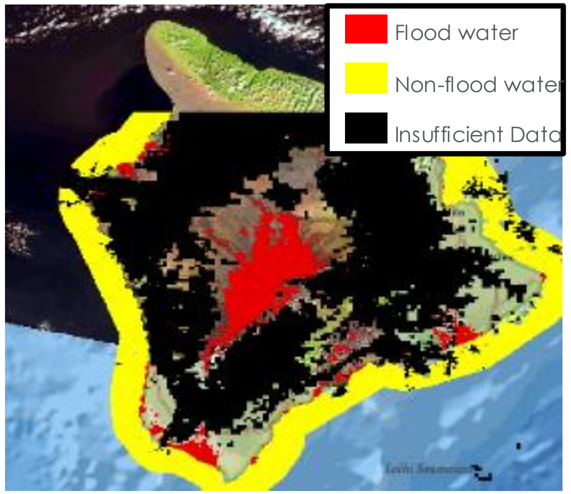
Landsat 8



MODIS (MCD12Q1) IGBP Land Cover



MODIS IGBP Land Cover with flood water

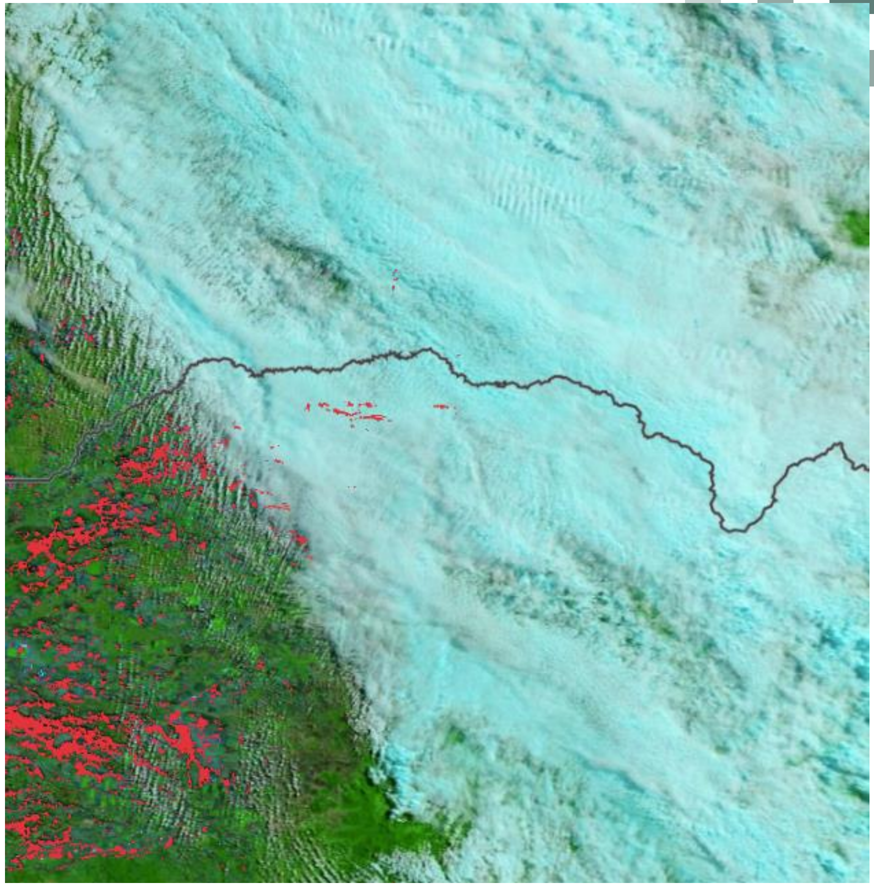


MODIS Flood Product

Limitations

Product Limitations Recap

- Cloudiness
  - False-negatives: can't see the flood
  - False-positives: cloud shadows detected as water
- 250 m resolution
  - Floods with limited spatial extent are difficult to pick up
- Twice Daily Observations (within 1-3 hours of each other)
  - Flash floods usually not captured, unless lucky with overpass timing
- Landcover limitations
  - Water under tree cover will not be visible to satellite
  - Urban – too many buildings/objects not under water
- Recent changes in normal surface water extent
  - New reservoirs (will be picked up by reference water layer after filled for 3 years)
  - Changes in river courses
  - ** Users can easily reclassify the product using a preferred surface water map, if available

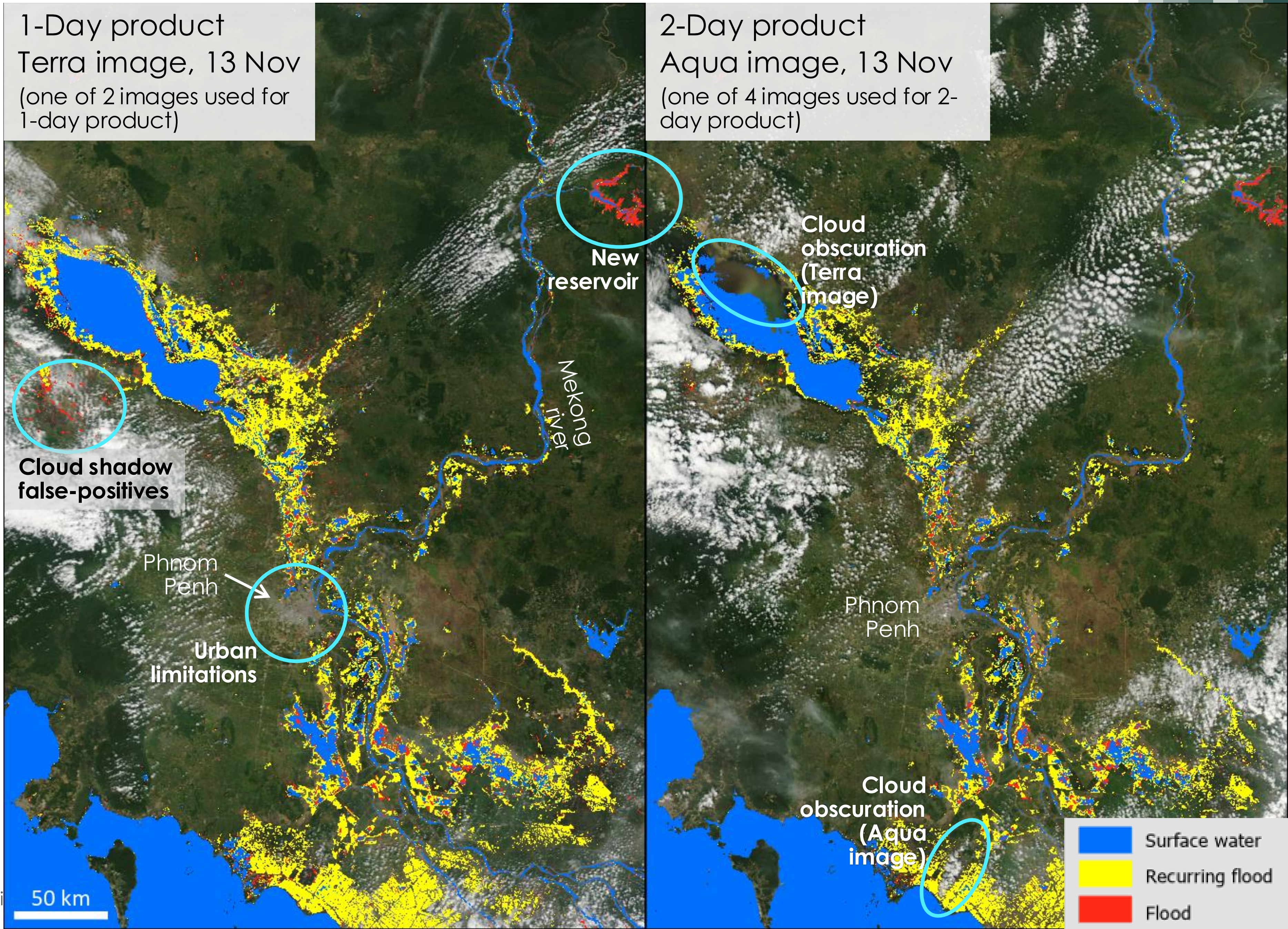


Limitations Example

13 Nov 2020, Cambodia

- Cloud obscuration / false-negatives
- Cloud false-positives (cloud shadows detected as water)
- Changes in normal surface water extent (new reservoir)
- Difficulties in urban areas

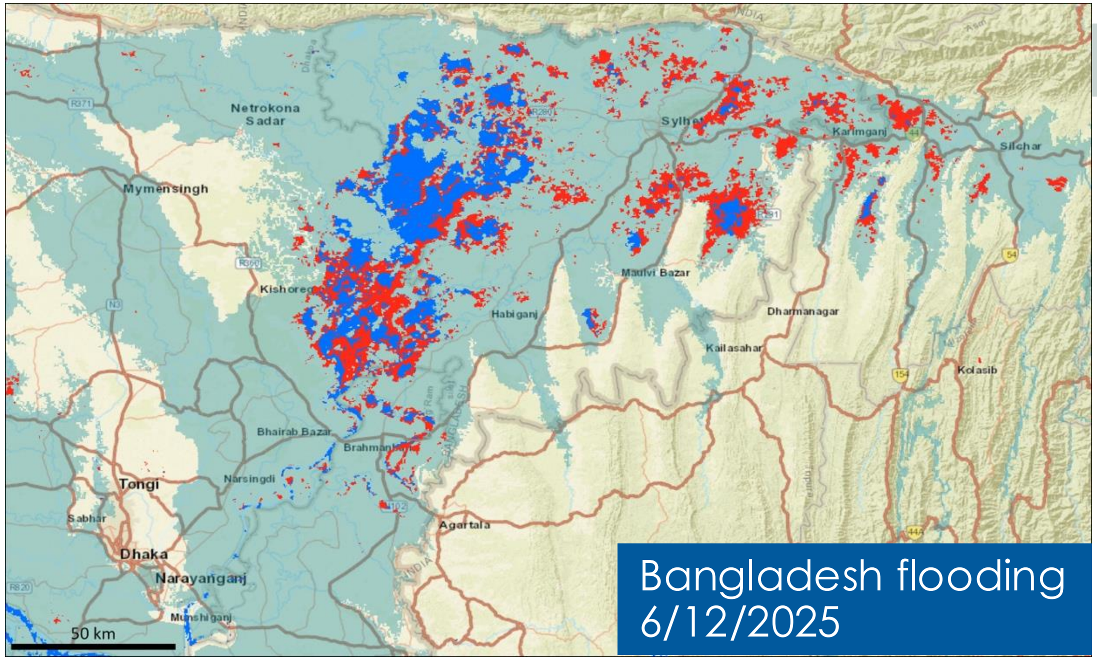
→ Looking at different products and contributing source images helps clarify which product is best for a given location



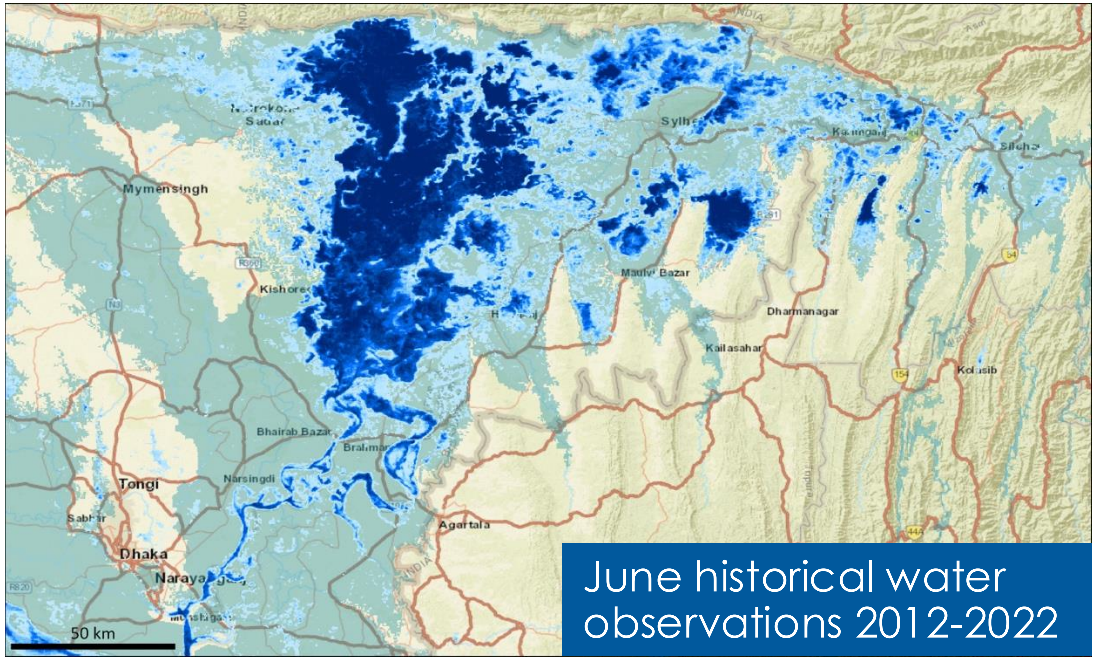
Archive

Product Archive

- Allows historical analyses and comparisons (such as our development of the recurring flood mask)
- Product fully reprocessed from Terra start of mission (early 2000) through 2025
    - 23 full years with Terra + Aqua: 2003-2025
    - Initial 2 years with just Terra (2000 - mid 2002)
    - Uses final geolocation and final surface reflectance, in place of NRT versions, which can occasionally have minor errors
- 2026 onwards: MODIS NRT products are being archived



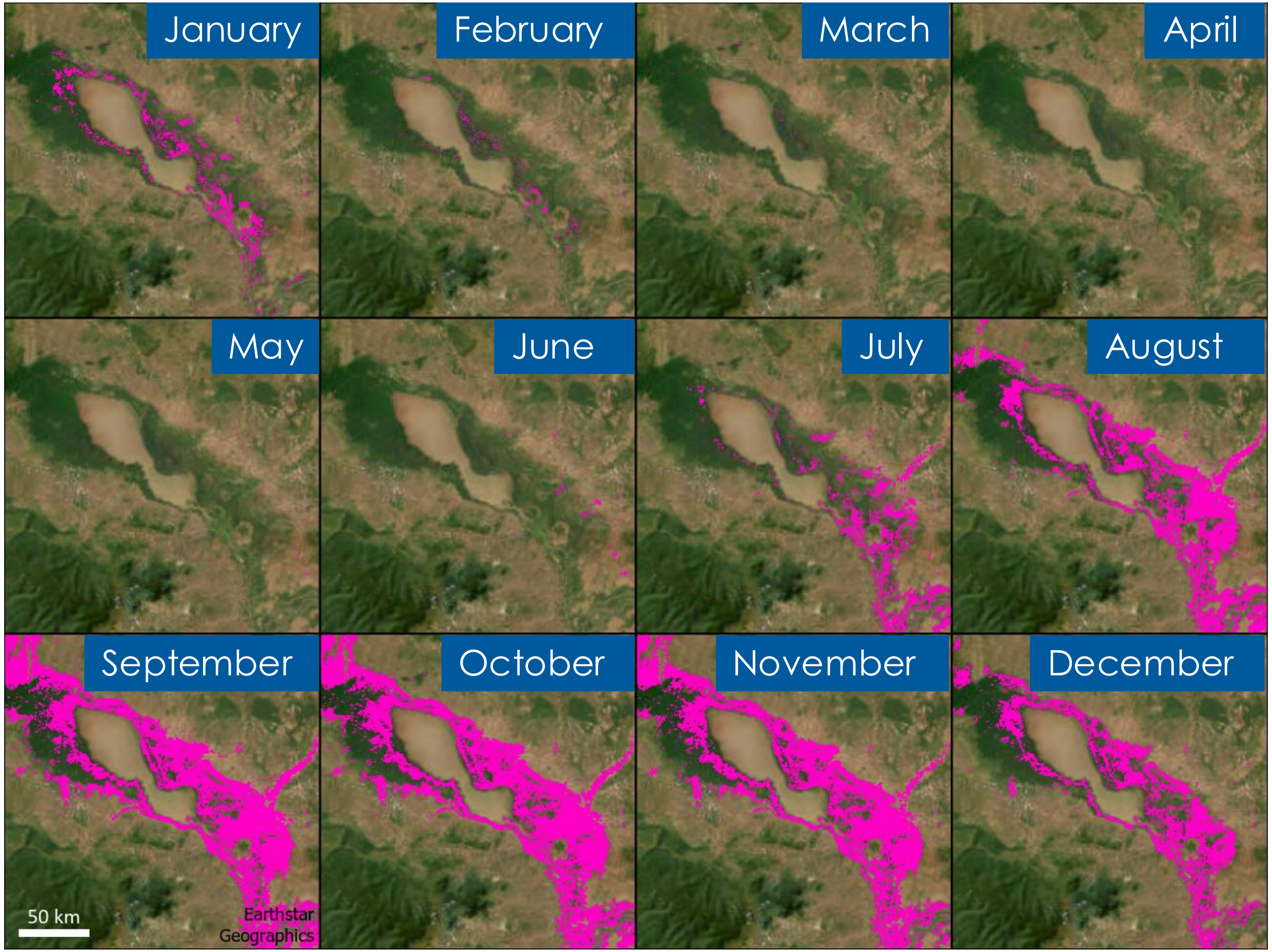
Bangladesh flooding 6/12/2025



June historical water observations 2012-2022

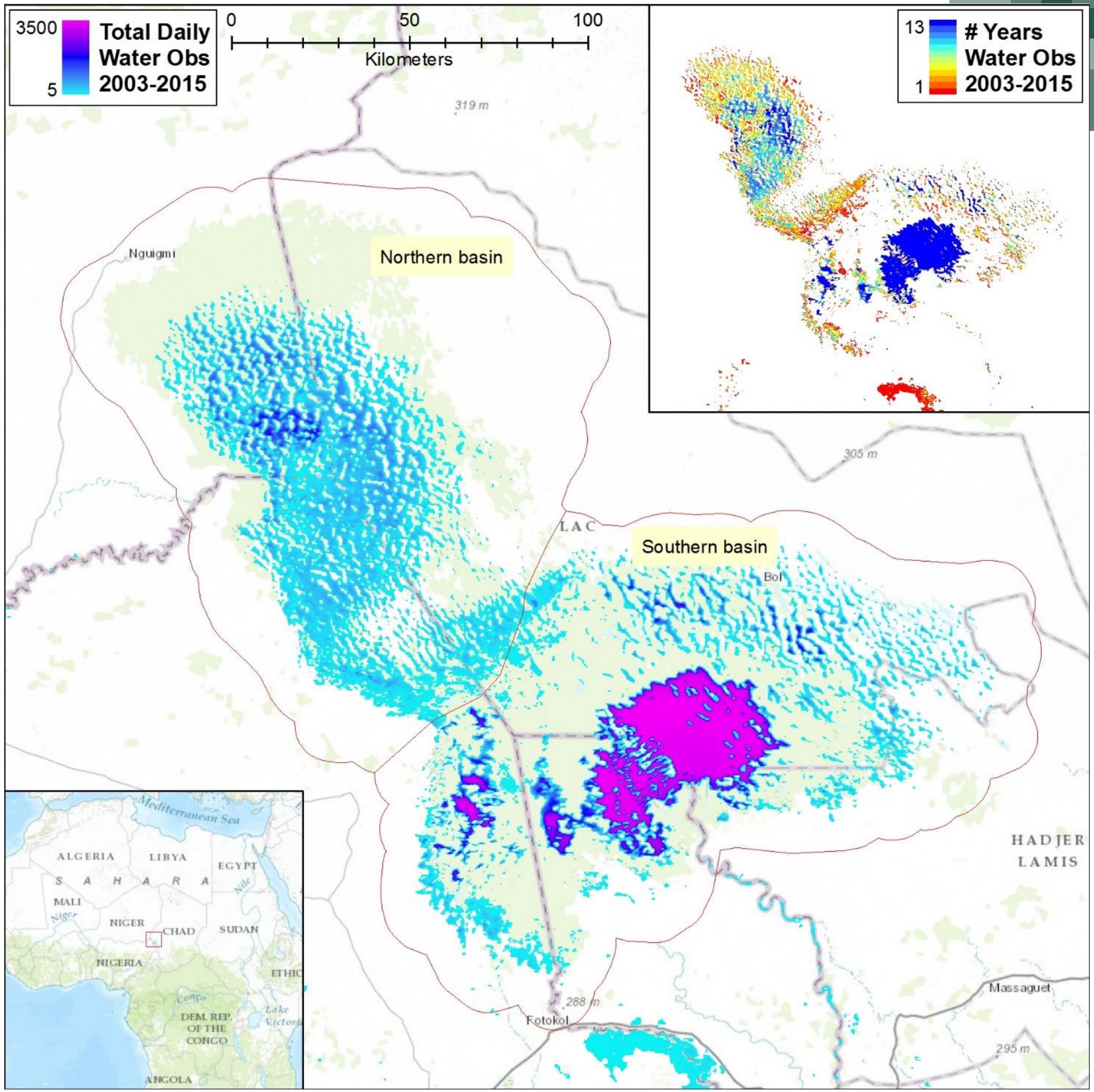
Recurring Flood: Derived From Archive

Cambodia / Tonle Sap lake: monthly recurring flood masks



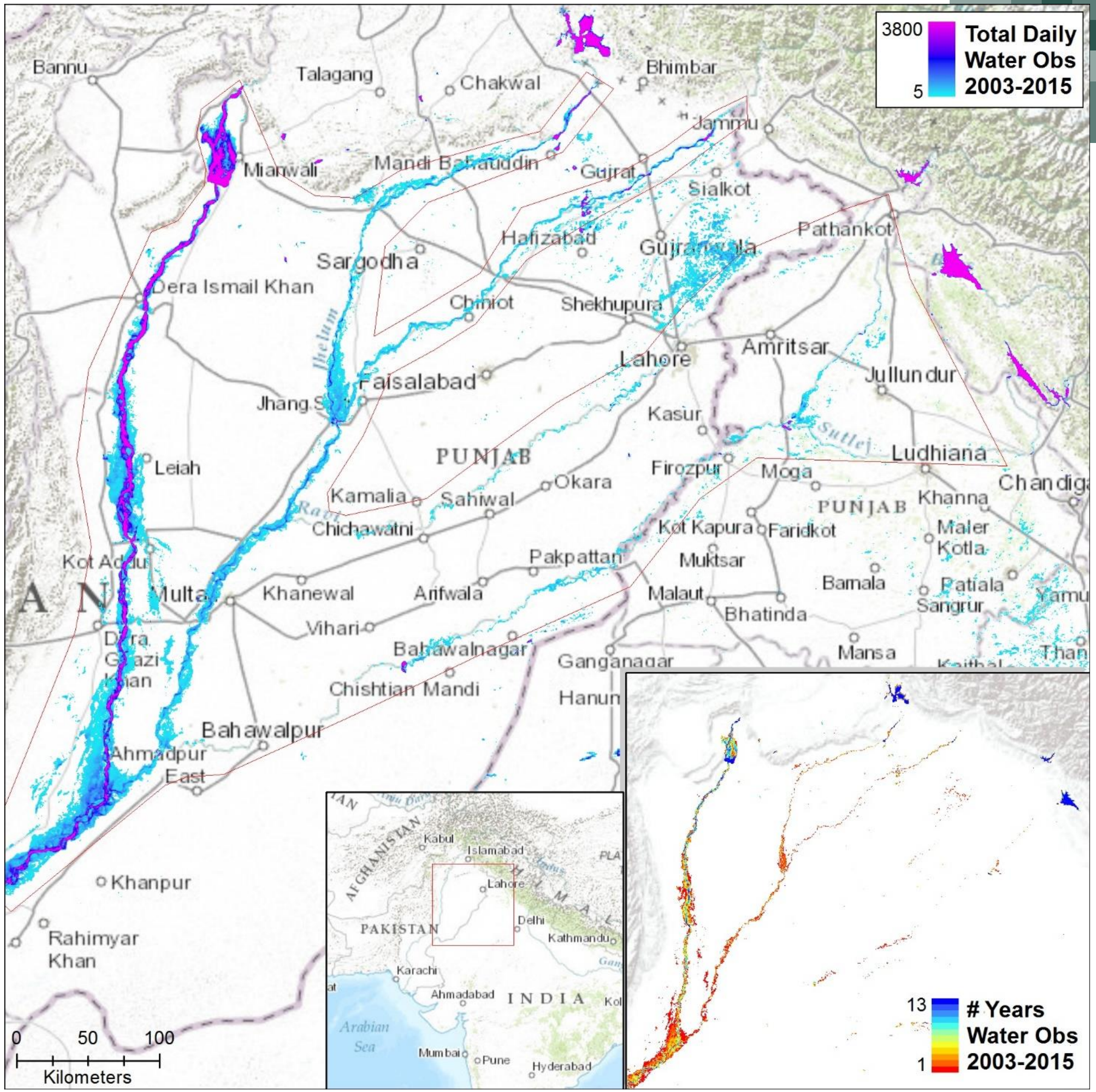
Historical Analysis
Lake Chad: 2003 – 2015

- Analysis of daily product shows how frequently different sections of the lake have held water over this period.
- Inset pane shows how many years (out of the 13 examined here) water was present.
- Note: this analysis ignored the flood/recurring flood/surface water distinction, and simply looked at any detected water, no matter how classified.



Historical Analysis
Indus 2003 – 2015

- Analysis of daily product shows how frequently the Indus river system has contained water (detected by this product) over 13 years, and its physical extent.
- **Note:** Some of the tributaries are not always detected, even when they likely contain water; this is a limitation of the 250 m pixel size.
- **Note:** this analysis ignored the flood/recurring flood/surface water distinction, and simply looked at any detected water, no matter how classified.

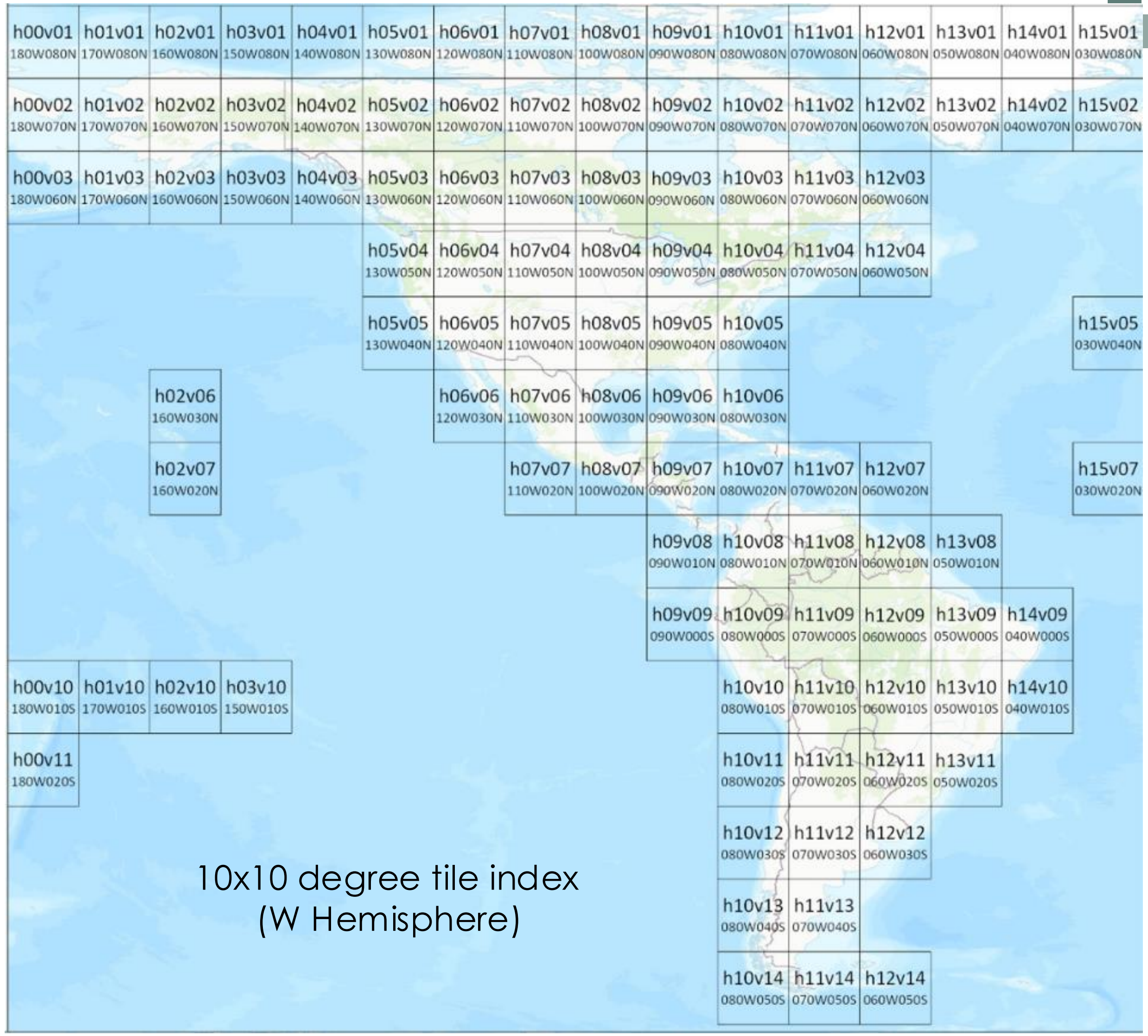


Distribution

Data Files

- Distributed in 10 x 10 degree tiles
  - 287 tiles globally in production
  - Full tile map on homepage, and in User Guide
- Core product: 1 HDF file / tile / day. Includes:
  - Flood layers (1, 2, 3-day composites).
  - Ancillary layers with: Water Counts, Valid Counts, and Total Counts. Allow users to create custom composites or examine underlying data.
  - Layers can be extracted to GeoTIFF files using GDAL tools (User Guide has examples/more info).
- Flood layer GeoTIFF files are also provided for NRT products
- The flood layer is a data layer (not RGB imagery) with values:

Flood layer pixel value	
0	No water
1	Surface water
2	Recurring flood
3	Flood
255	Insufficient data

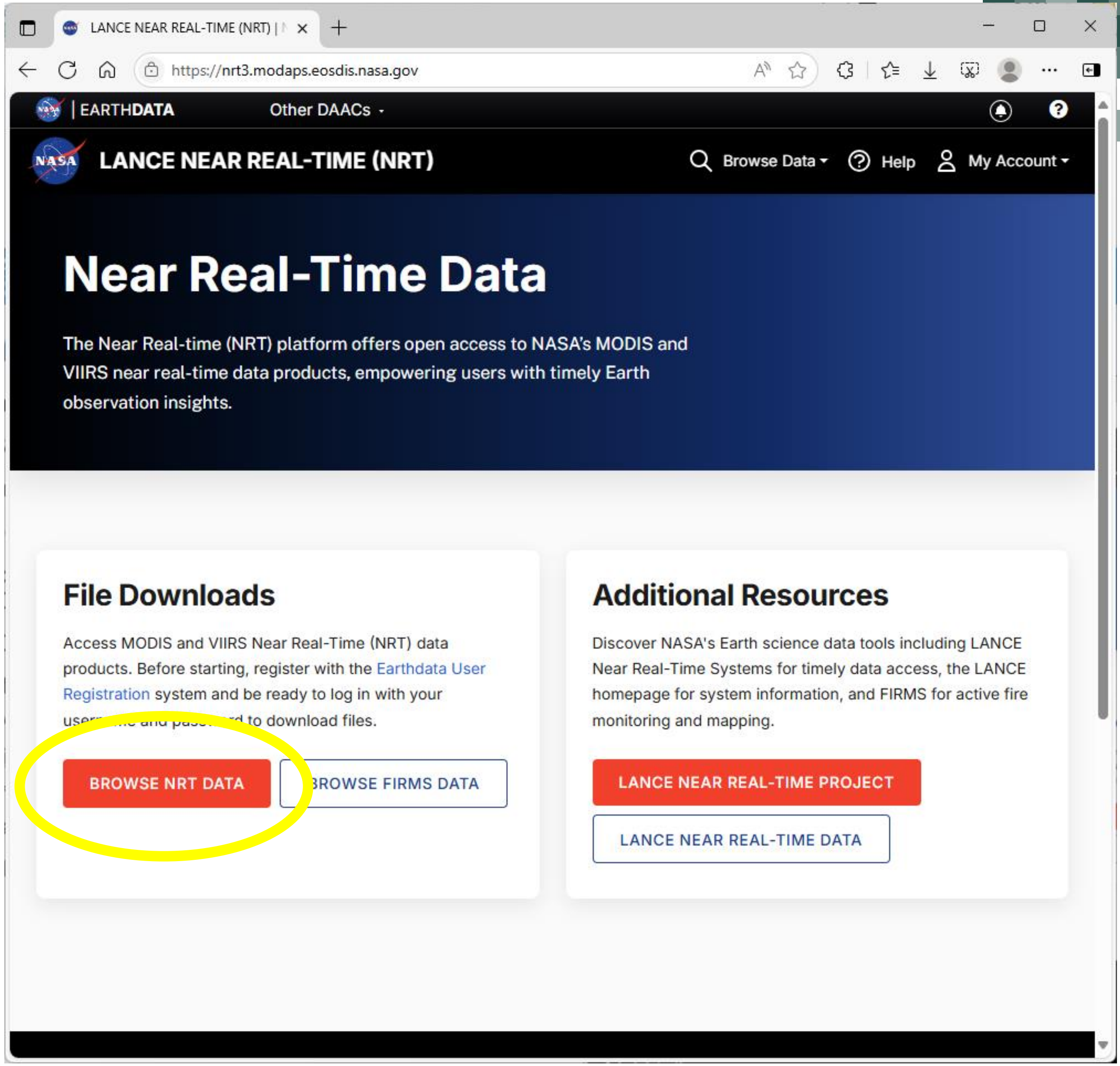


10x10 degree tile index
(W Hemisphere)

NRT Product Downloads

- LANCE NRT download sites (and nrt4 backup):
  - Only retain data for ~ 1 week
  - Free NASA EarthData account is required
- Navigate: "Browse NRT Data" → allData → <COLLECTION> → <PRODUCT> → <YEAR> → <DOY> (day-of-year)
- <COLLECTION> = 61 for MODIS product
- <COLLECTION> = 5200 for VIIRS product
- <PRODUCT> for HDF files:
  - For MODIS: MCDWD_L3_NRT
  - For VIIRS: VCDWD_L3_NRT
  - For GeoTIFF files, F1, F2, or F3 are inserted to indicate which composite: eg MCDWD_L3_F2_NRT for 2-day product.

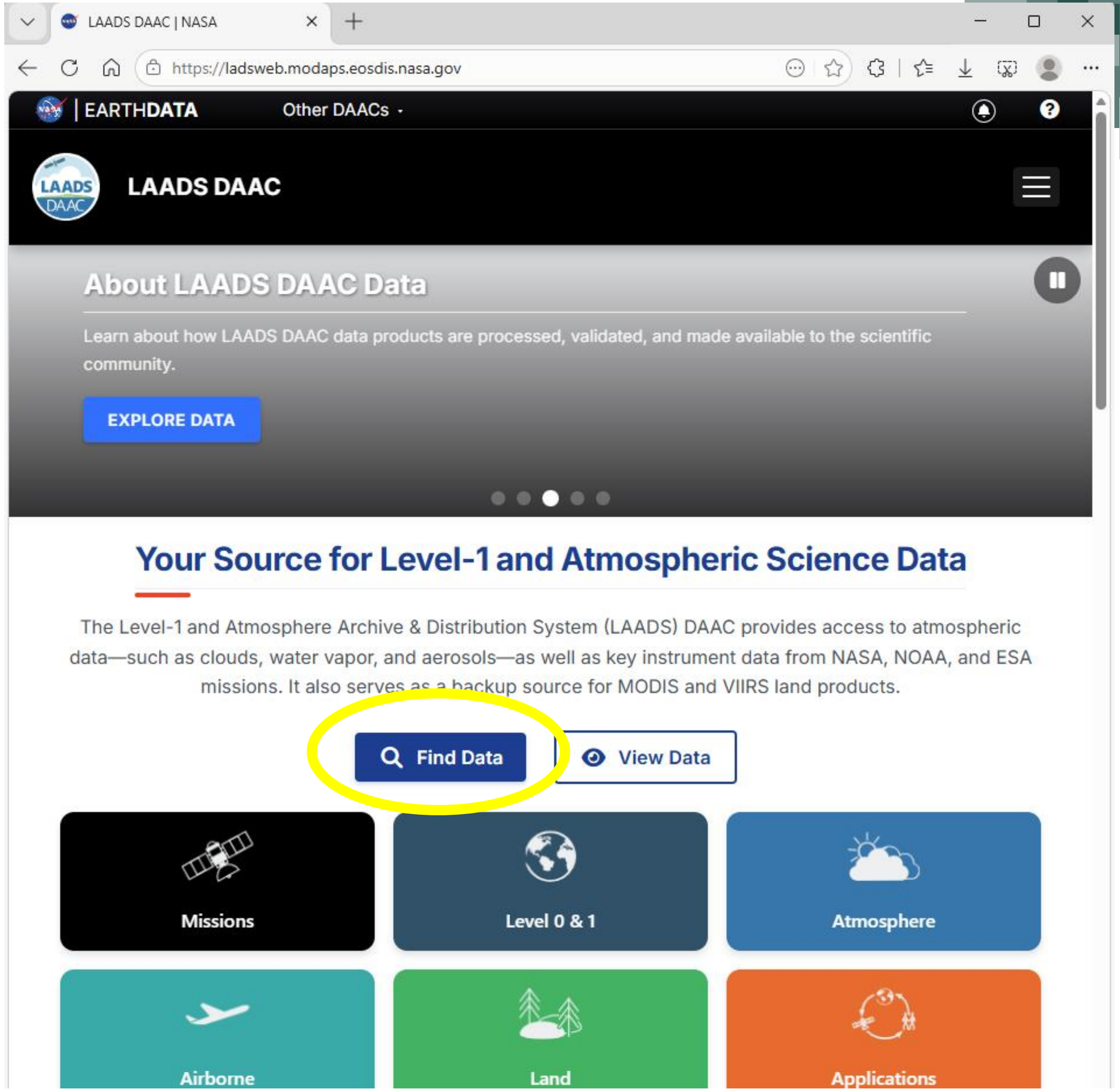
Source	Shortname	NRT product	Archive product
MODIS	MCDWD	MCDWD_L3_NRT	MCDWD_L3
VIIRS	VCDWD	VCDWD_L3_NRT	



Archived Product Downloads

- Currently only MCDWD products, both reprocessed (2000-2025) and NRT (2026 onwards only) are archived
- LAADS DAAC
- Free NASA EarthData account required to download
- Navigate: “Find Data” → Online Archive → <COLLECTION> → <PRODUCT> → <YEAR> → <DOY> (day-of-year)

Source	Shortname	NRT product	Archive product
MODIS	MCDWD	MCDWD_L3_NRT	MCDWD_L3
VIIRS	VCDWD	VCDWD_L3_NRT	



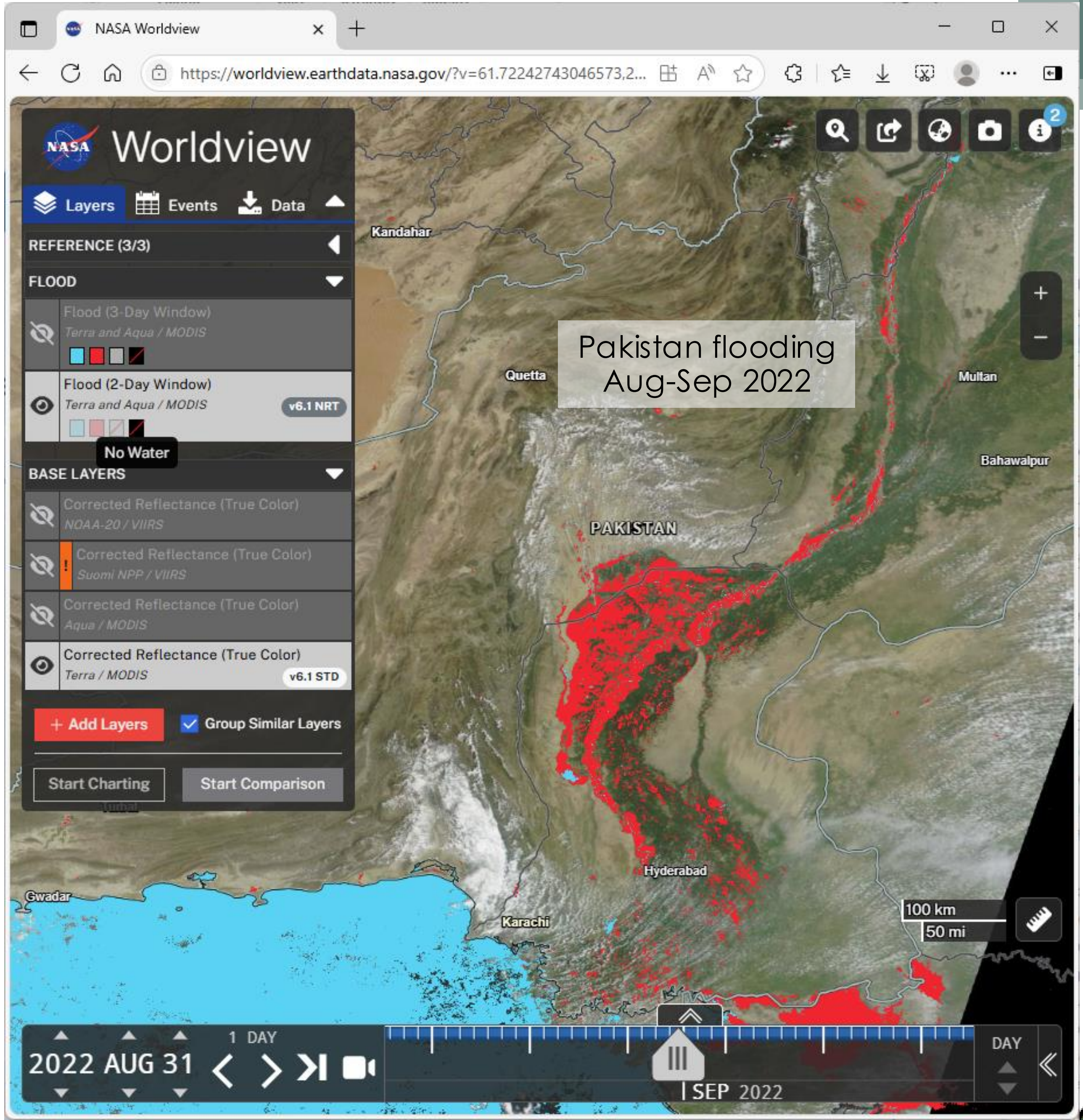
Data Access: APIs & Worldview

NASA GIBS (Global Imagery Browse Service)

- WMS, WMTS, TWMS services providing imagery, not data
- Accessible via: ArcGIS, QGIS, Google Earth, NASA Worldview
- Flood plus 1000+ other satellite imagery products

Browsable web application: [NASA Worldview]

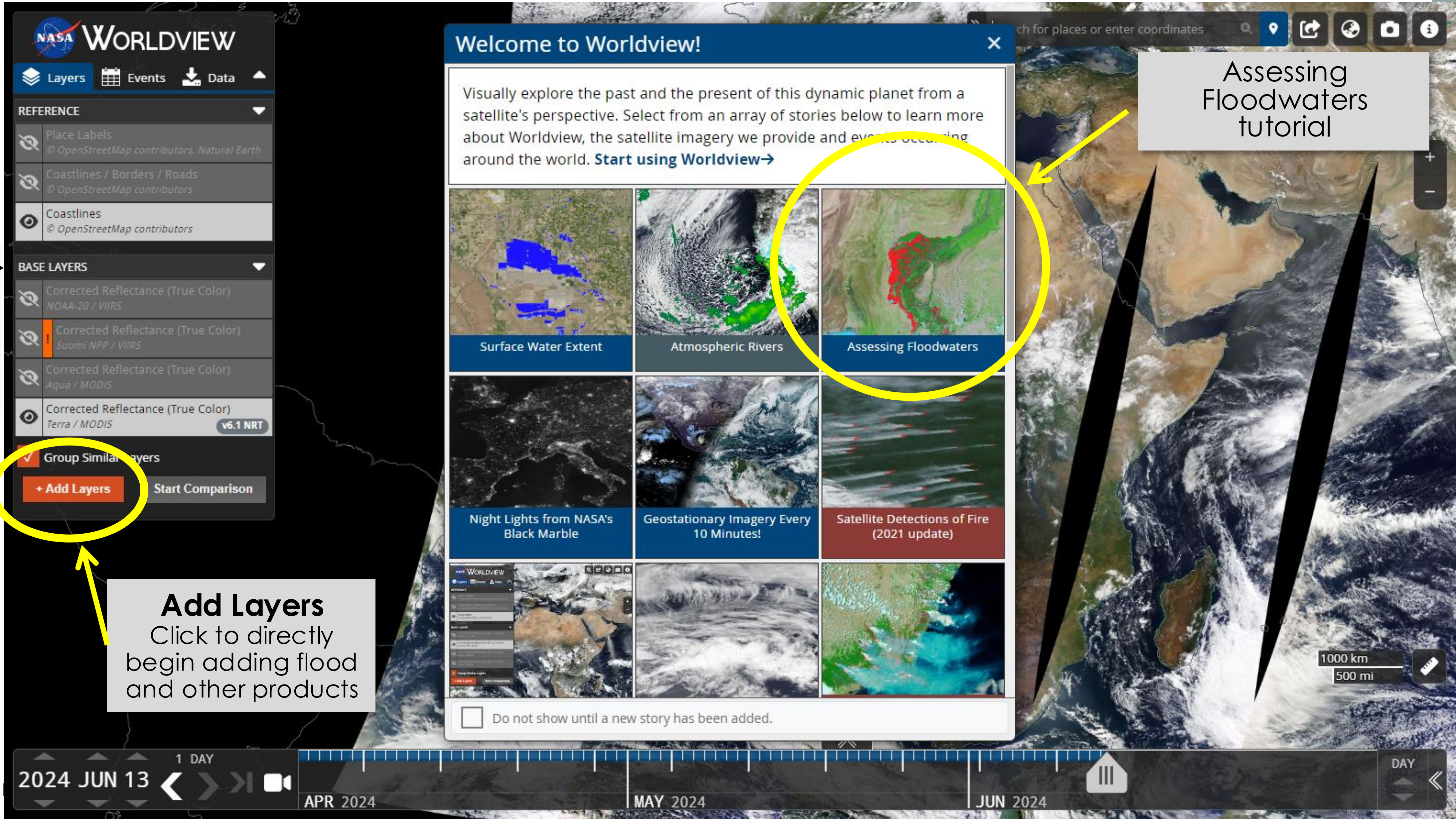
- MODIS Flood products from early 2021
- VIIRS Flood products from June 2025
- + Many other satellite imagery products (1000's)
- Allows comparison with source imagery
- Allows comparison between dates or products
- Only hosts 2 and 3-day flood products (not 1-day)
- Navigate to: "Add Layers" → Floods → Flood
  → "Terra and Aqua MODIS"
  or → "NOAA-20 and NOAA-21/VIIRS"
- [Worldview pre-loaded with MODIS and VIIRS flood products]
- Tutorial "Assessing Floodwaters" available on Worldview start page



Data Access: Worldview

[Access Worldview]

Base Layers
Pre-loaded, includes MODIS and VIIRS imagery

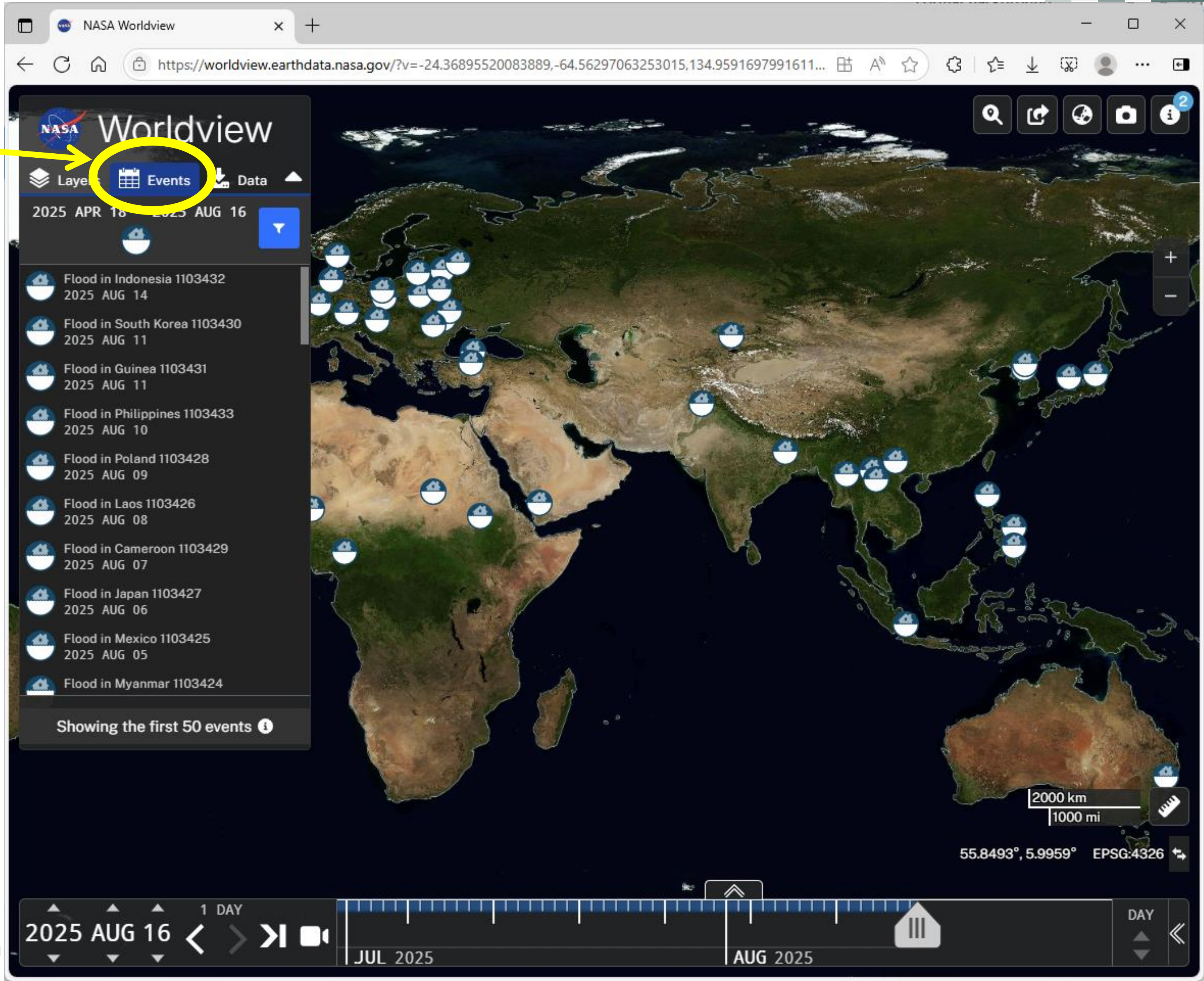


Assessing Floodwaters tutorial

Add Layers
Click to directly begin adding flood and other products

Data Access: Worldview

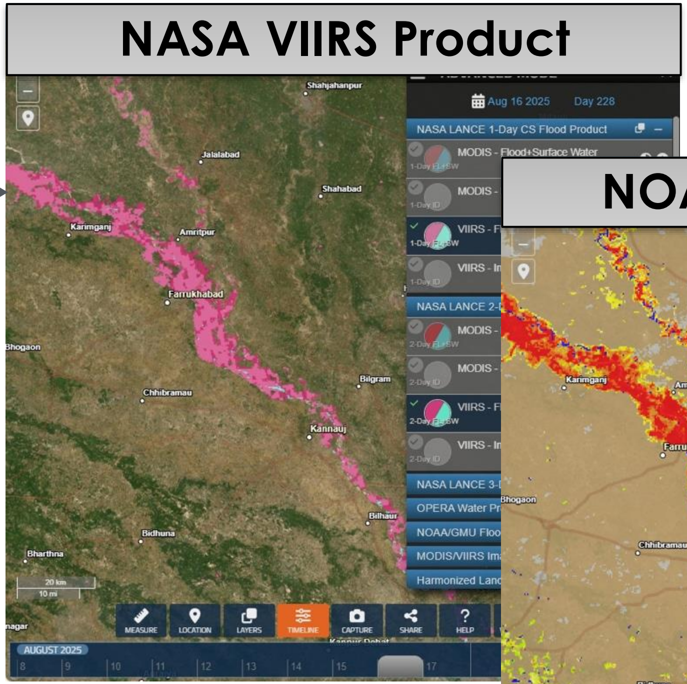
- New Feature: Flood Events
- Lists recent events, with locations, and dates
- Can then compare to flood products available in Worldview
- Source: Global Disaster Alert and Coordination System (UN/EC)

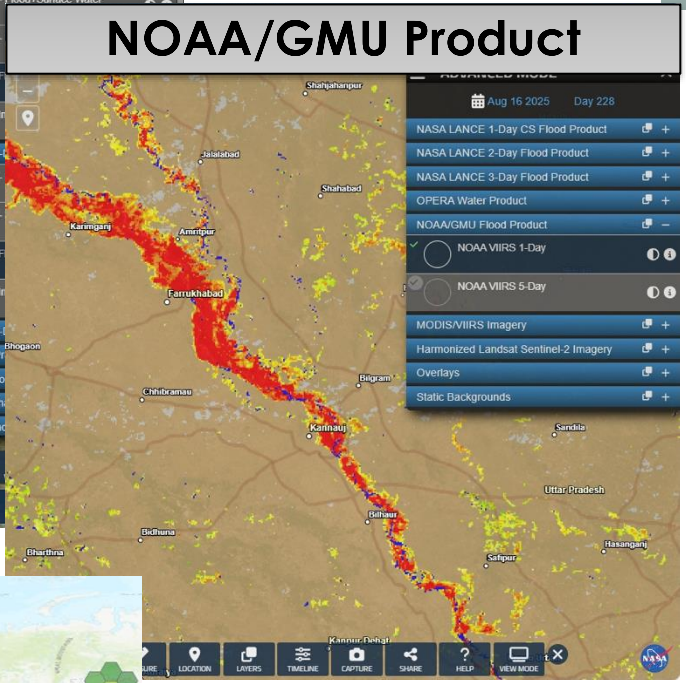


Future Directions

Future Directions

- Release of FLOOD viewer
    - Adds 1-day MODIS and VIIRS products
    - Adds non-NASA flood products
- Archiving of VIIRS NRT products
- Machine learning update to water detection algorithm (VIIRS)
    - Significantly improved water detection detail
    - Rarely detects shadows as water
- Global alerts to highlight active flooding at a global scale
- MODIS flood product archive available in GIBS/Worldview (currently only 2021 forward)
- Sentinel-3/OLCI product
    - Less-cloudy AM observations to improve flood detection





Global Flood Product: Data Access and Worldview Demo

Resources

- Global Flood Product Homepage
  - User Guide & FAQs, mailing list
  - Links to LANCE download sites
- Worldview
- Worldview with flood loaded
- Questions/Suggestions: dan.slayback@nasa.gov or earthdata-support@nasa.gov

Product Homepage

Part 1:
Summary

Global Flood Products

- Description of Global Flood Product development approach:
  - Based on Terra & Aqua MODIS, NOAA-20 and -21 VIIIRS Imagery
  - Uses red, infrared, and shortwave infrared reflectance to detect water
  - Multi-look composites (1, 2, 3 days) to assign threshold-based water detection
  - Terrain shadow and cloud shadow correction to remove false positive in water detection

- Selection of appropriate composite product based on checking true color imagery for cloud cover

- Identification of recurring floods based on 22-year historical flood mask data

- Examples and demonstration of flood cases using Worldview
  - Recent flood cases in Russia, Australia, and Mozambique

- Flood data access and download using Worldview, LAADS DAAC
  - Near-real time and 23 years of archived flood data from Terra + Aqua: 2003-2025
  - Initial 2 years from Terra (2000 - mid 2002) Flood data can be downloaded as GeoTIFF
  - Recent data (2021to present) available from Worldview
  - Flood data can be downloaded as GeoTIFF and analyzed in QGIS

Limitations and Future Plans

Limitations:

- Cannot see floods through clouds
- Cloud shadow detected as water
- Medium spatial resolution (250 m), may not be adequate for urban floods
- Based on twice daily observations: may miss flash floods
- Water under tree cover not visible to satellite
- Changes in surface water extent: new reservoirs, changes in river courses may not be in reference surface water mask

Future Plans:

- Release of new FLOOD viewer: Include NOAA-GMU and Copernicus-GLoFAS products
- Archive VIIRS products
- Use Machine learning update to water detection algorithm (VIIRS)
- Include global alerts to highlight active flooding at a global scale
- Include Sentinel-3/OLCI product

Looking Ahead to Part 2

Monitoring Floods using OPERA Surface Water Extent based on Optical and SAR Observations.

Homework and Certificates

- **Homework:**
  - One homework assignment
  - Opens on 25/06/2026
  - Access from the training webpage
  - Answers must be submitted via Google Forms
  - **Due by 09/07/2026**

- **Certificate of Completion:**
  - Attend all live webinars (attendance is recorded automatically)
  - Complete the homework assignment by the deadline
  - You will receive a certificate via email approximately two months after completion of the course.

Acknowledgements

Dr. Daniel Slayback

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618 NASA-GSFC

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  - Sean.mccartney@nasa.gov
- Erika Podest
  - Erika.podest@jpl.nasa.gov

- ARSET Website
- ARSET YouTube

For questions, comments, or to share how you have applied our trainings to your work or studies, email nasa.arset@gmail.com.

Join our mailing list to stay up-to-date on our latest trainings. Visit our Contact page to subscribe.

Resources

- Global Flood Product
  - User Guide & FAQs, mailing list
  - Links to LANCE download sites
- Worldview
- Worldview with flood loaded
- Questions/Suggestions: dan.slayback@nasa.gov or earthdata-support@nasa.gov

Product homepage

Reference: Jonkman, S.N., Curran, A. & Bouwer, L.M. Floods have become less deadly: an analysis of global flood fatalities 1975–2022. Nat Hazards 120, 6327–6342 (2024). https://doi.org/10.1007/s11069-024-06444-0

Thank You!